
Computer Networks
DS360
Project-Phase 1

Deadline: Day 26/07/2026 @ 23:59

[Total Mark is 5]

Student Details:

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Instructions:

- You must submit two separate copies (**one Word file and one PDF file**) using the Assignment Template on Blackboard via the allocated folder. These files **must not be in compressed format**.
- It is your responsibility to check and make sure that you have uploaded both the correct files.
- Zero mark will be given if you try to bypass the SafeAssign (e.g., misspell words, remove spaces between words, hide characters, use different character sets, **convert text into image** or languages other than English or any kind of manipulation).
- Email submission will not be accepted.
- You are advised to make your work clear and well-presented. This includes filling your information on the cover page.
- You must use this template, failing which will result in zero mark.
- You **MUST** show all your work, and text must not be converted into an image, unless specified otherwise by the question.
- Late submission will result in ZERO mark.
- The work should be your own, copying from students or other resources will result in ZERO mark.
- Use **Times New Roman** font for all your answers.

Project Instruction

Project Objective:

Design, configure, and test a network setup in **Packet Tracer** to successfully transmit data packets between devices in different LANs across two separate WANs (WAN 1 and WAN 2).

Group Details

- Group Size = Maximum of 3 Members.
- One group member (group leader/coordinator) must **submit the project report** on blackboard. Marks will be given based on your submission and quality of the contents.

Project Report

- The project is divided into two phases:
 - Phase 1 (Mid project Progress Report)
 - Phase 2 (Final project Report)
- This report is for Phase 1. In this report, you should complete all the requirements for Phase 1 only.
- Each Report will be evaluated according to the marking criteria

Project Description (Common to Phase 1 and Phase 2)

A university operates two campuses: a Main Campus and a Branch Campus. Each campus network is treated as a separate WAN site. Your task is to design and configure, in Cisco Packet Tracer, a network that successfully transmits data packets from a device in any LAN of WAN 1 (Main Campus) to a device in any LAN of WAN 2 (Branch Campus). Each WAN contains 2 LANs, and each LAN has 3 end devices.

Network Design Requirements:

1. WAN 1 (Main Campus):

- Contains LAN 1 (Computer Lab) and LAN 2 (Admin Office).
- Each LAN has 3 devices (PCs or laptops) connected through a switch.
- Assign private IP addresses to the devices in each LAN (e.g., use 172.16.10.0/24 for LAN 1 and 172.16.20.0/24 for LAN 2).
- Connect each LAN to a router (Router 1).

2. WAN 2 (Branch Campus):

- Contains LAN 3 (Library) and LAN 4 (Faculty Office).
- Each LAN has 3 devices (PCs or laptops) connected through a switch.
- Assign private IP addresses to the devices in each LAN (e.g., use 172.16.30.0/24 for LAN 3 and 172.16.40.0/24 for LAN 4).
- Connect each LAN to a router (Router 2).

3. WAN Connection:

- Connect Router 1 and Router 2 using a serial link to simulate a WAN connection.
- Assign IP addresses to the serial interfaces (e.g., 10.10.10.1/30 for Router 1 and 10.10.10.2/30 for Router 2).

Configuration Details:

1. Assign IP Addresses:

- Configure IP addresses, subnet masks, and default gateways for all devices in each LAN. Example:
 - LAN 1: 172.16.10.0/24, Default Gateway: 172.16.10.1 (Router 1's interface).
 - LAN 2: 172.16.20.0/24, Default Gateway: 172.16.20.1 (Router 1's interface).
 - LAN 3: 172.16.30.0/24, Default Gateway: 172.16.30.1 (Router 2's interface).

- LAN 4: 172.16.40.0/24, Default Gateway: 172.16.40.1 (Router 2's interface).

2. Router Configuration:

- Configure IP addresses on the LAN and WAN interfaces of both routers. Example:
 - Router 1:
 - LAN 1 Interface: 172.16.10.1/24
 - LAN 2 Interface: 172.16.20.1/24
 - WAN Interface (Serial): 10.10.10.1/30
 - Router 2:
 - LAN 3 Interface: 172.16.30.1/24
 - LAN 4 Interface: 172.16.40.1/24
 - WAN Interface (Serial): 10.10.10.2/30

3. Enable Routing:

- Configure static routes or a dynamic routing protocol (e.g., RIP) on both routers to ensure communication between WAN 1 and WAN 2.
-

Phase 1 Requirements**(5 Marks)**

Students are required to submit a short mid progress report, which must include the following:

1. **Work completed this period (2 marks):** This includes the installation of Cisco Packet Tracer, and a study of Packet Tracer for creating nodes, using different types of links (copper straight-through, copper cross-over, serial DCE/DTE), and assigning IP addresses.
 - **Note:** You must show a screenshot of the installation of Packet Tracer (with your system visible, e.g., date/time or username). (1 mark)
 - Give a description of the objectives and advantages of using Packet Tracer for this project. (1 mark)
2. **One key decision made during this period (1 mark):** For example, your choice of IP addressing scheme, choice of routing method to study, or division of work among group members. Explain briefly why the decision was made.
3. **One problem encountered (1 mark):** Describe one genuine problem faced so far (technical or organizational) and its current status (resolved / in progress).
4. **Plan for the next period (0.5 mark):** Briefly list the tasks you plan to complete before the final submission (e.g., building the topology, configuring routers, testing with ping).
5. **Each group member's input/participation (0.5 mark):** State each member's name and their specific contribution during this period.

Each Group Member's Input / Participation

- **Dalal Ali (Team Leader):** Installed Cisco Packet Tracer, prepared the IP addressing scheme, and configured Router 1 with its LAN interfaces and WAN serial interface.
- **Raghad Hussain:** Prepared the project report, organized and formatted the Word document, and documented the network design and IP addressing tables.
- **Misaa Sultan:** Configured Router 2, verified the network configuration, and uploaded the Packet Tracer (.pkt) file to the GitHub repository.
- **Saba Alharbi:** Assisted in documenting the router configuration steps, organized the screenshots, and formatted the project report.
- **Reem Alharbi:** Prepared the Reflection section, reviewed the report, and verified the project links before submission.

Version History / Commit Log Requirements (Common to Both Reports)

Students must document and submit their project history/timestamps using a live, editable link:

- **Written report:** Submit a shared Google Doc or Word Online link (sharing set to "Anyone with the link can view") along with your final PDF report. The link must show the progress of the students' work with timestamps/screenshots.
- **Code/data/configuration files:** Submit Packet Tracer files (.pkt) and any configuration scripts via a GitHub or GitLab repository link (the coordinator may choose a repository suitable for the course). The repository must show a commit history reflecting incremental progress. Commit regularly (at minimum once per week), with clear, descriptive commit messages.

A separate section in the report must be added and named "Project Artifacts" to include all links, for example:

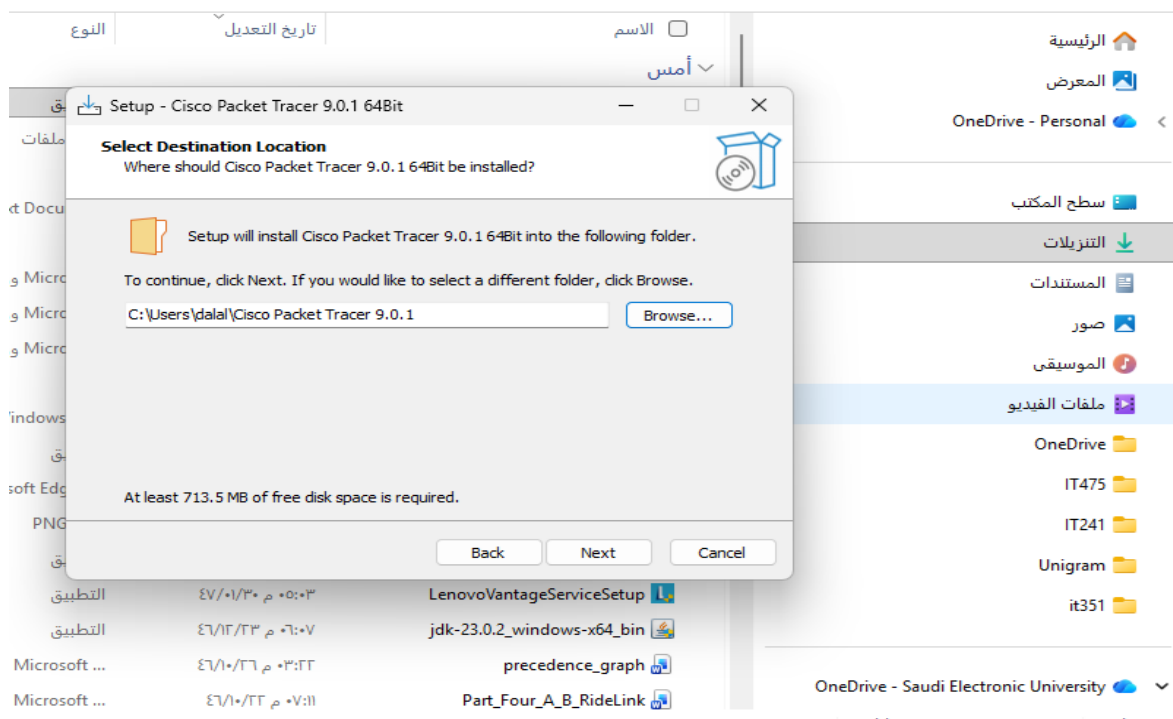
- Live report document (with edit history): [link]
- GitHub/GitLab repository (with commit history): [link]
- Packet Tracer project file (.pkt): [link or attached]

What NOT to do: Students must not write their report elsewhere and paste the final version in at the last minute. They also must not make a single commit containing the entire project. Both are treated as red flags regardless of the quality of the final work.

Important Notes

- **AI use:** Students may use AI tools (ChatGPT, Claude, Copilot, Gemini, etc.) only for brainstorming approaches, debugging small errors, or proofreading text. AI must not be used to generate analysis, write configuration logic, or produce substantial portions of the report or project files. Any permitted AI use must be disclosed in the Reflection section. Undisclosed AI use, or AI use beyond this scope (disclosed or not), will result in a zero grade for the entire project.
- Any submission lacking a valid, inspectable version/commit history, or showing evidence of AI generation or plagiarism, will receive a zero grade for the entire project.
- Failure to submit the mid progress report as required, or submission of reports found to be fabricated, copied, or AI-generated, will result in a zero grade for the entire project.
- The mid progress report should be concise including screenshots.
- The same shared document link and repository submitted in Phase 1 must continue to be used for Phase 2, so that the full project history is visible in one place.

Work completed this period (2 marks): the installation of Cisco Packet Tracer,



Answer:**IP Addressing Scheme****LAN 1 – Computer Lab**

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1-Main	GigabitEthernet0/0	172.16.10.1	255.255.255.0	—
Lab-PC1	FastEthernet0	172.16.10.11	255.255.255.0	172.16.10.1
Lab-PC2	FastEthernet0	172.16.10.12	255.255.255.0	172.16.10.1
Lab-PC3	FastEthernet0	172.16.10.13	255.255.255.0	172.16.10.1

Network Address: 172.16.10.0/24**Broadcast Address:** 172.16.10.255**LAN 2 – Admin Office**

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1-Main	GigabitEthernet0/1	172.16.20.1	255.255.255.0	—
Admin-PC1	FastEthernet0	172.16.20.11	255.255.255.0	172.16.20.1
Admin-PC2	FastEthernet0	172.16.20.12	255.255.255.0	172.16.20.1
Admin-PC3	FastEthernet0	172.16.20.13	255.255.255.0	172.16.20.1

Network Address: 172.16.20.0/24**Broadcast Address:** 172.16.20.255

LAN 3 – Library

Device	Interface	IP Address	Subnet Mask	Default Gateway
R2-Branch	GigabitEthernet0/0	172.16.30.1	255.255.255.0	—
Library-PC1	FastEthernet0	172.16.30.11	255.255.255.0	172.16.30.1
Library-PC2	FastEthernet0	172.16.30.12	255.255.255.0	172.16.30.1
Library-PC3	FastEthernet0	172.16.30.13	255.255.255.0	172.16.30.1

Network Address: 172.16.30.0/24

Broadcast Address: 172.16.30.255

LAN 4 – Faculty Office

Device	Interface	IP Address	Subnet Mask	Default Gateway
R2-Branch	GigabitEthernet0/1	172.16.40.1	255.255.255.0	—
Faculty-PC1	FastEthernet0	172.16.40.11	255.255.255.0	172.16.40.1
Faculty-PC2	FastEthernet0	172.16.40.12	255.255.255.0	172.16.40.1
Faculty-PC3	FastEthernet0	172.16.40.13	255.255.255.0	172.16.40.1

Network Address: 172.16.40.0/24

Broadcast Address: 172.16.40.255

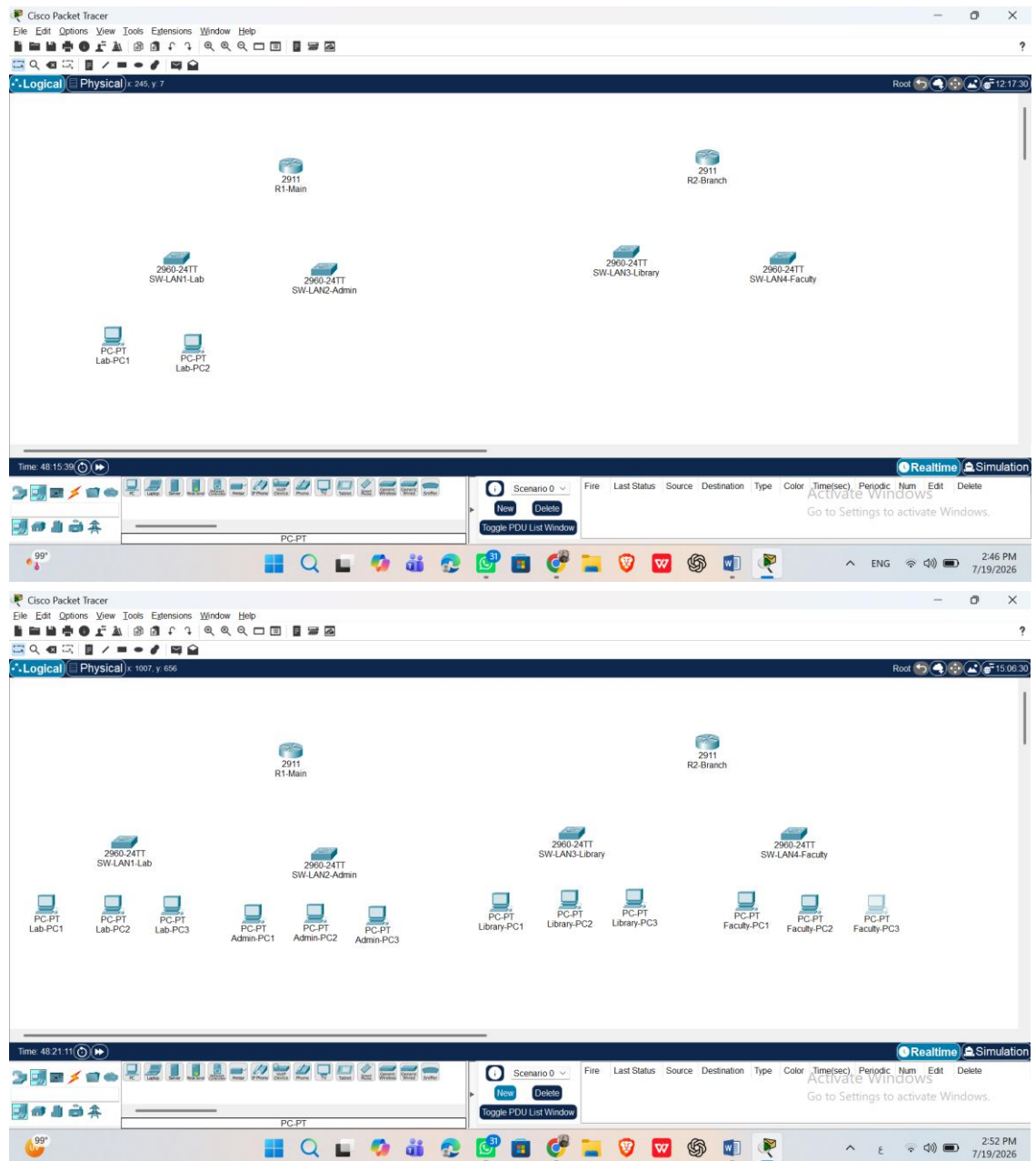
WAN Serial Link

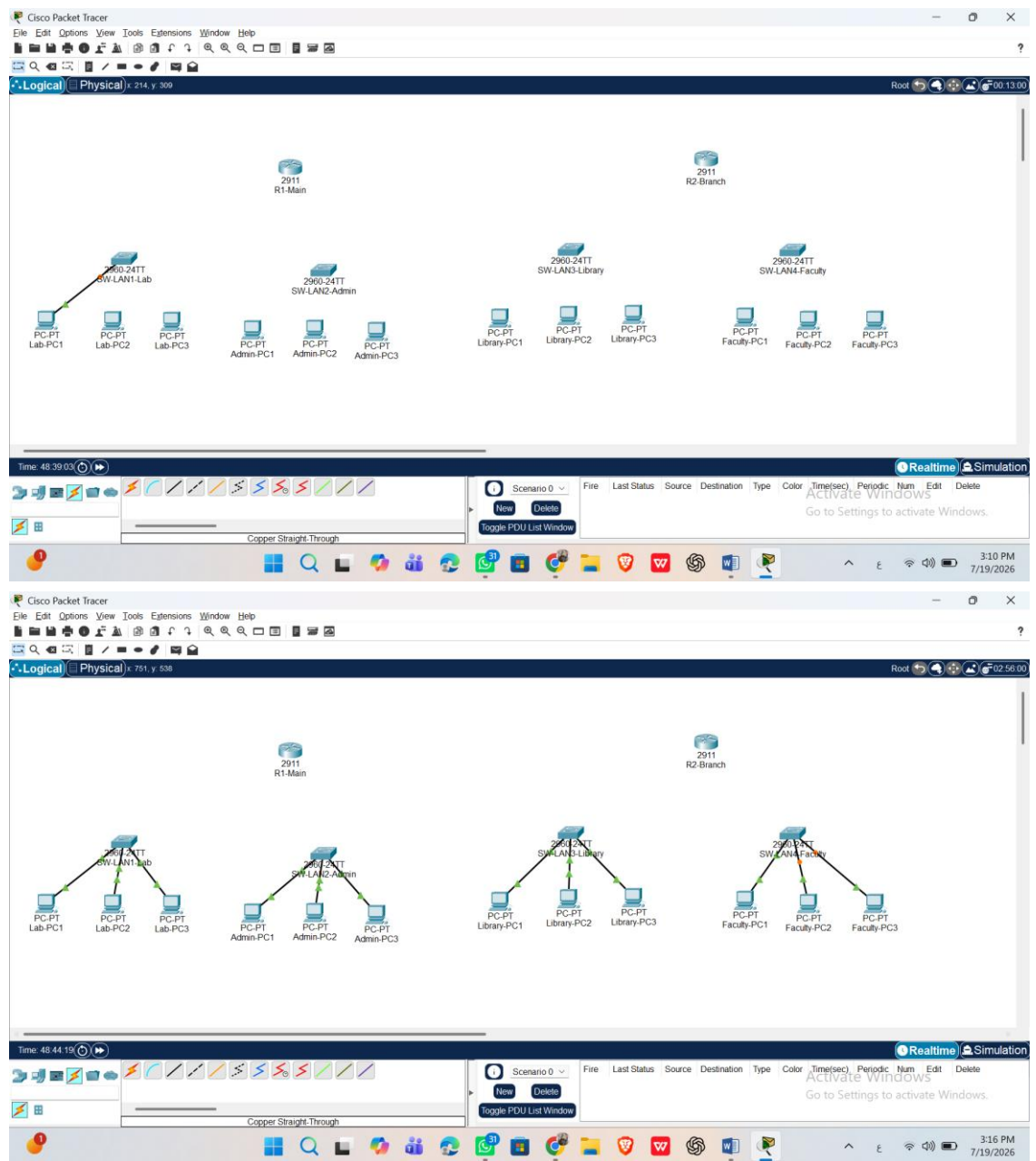
Device	Interface	IP Address	Subnet Mask
R1-Main	Serial0/3/0	10.10.10.1	255.255.255.252
R2-Branch	Serial0/3/0	10.10.10.2	255.255.255.252

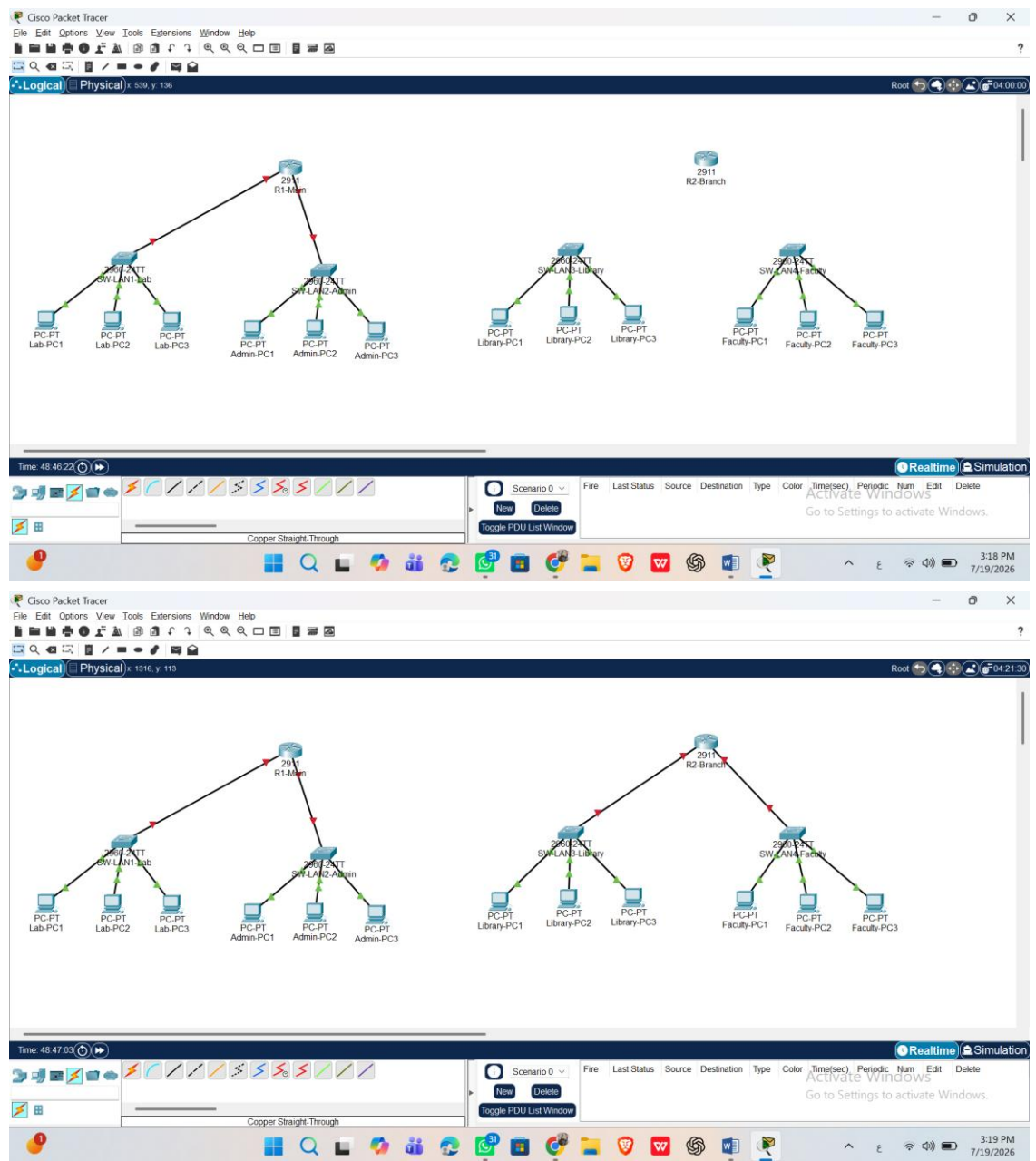
Network Address: 10.10.10.0/30

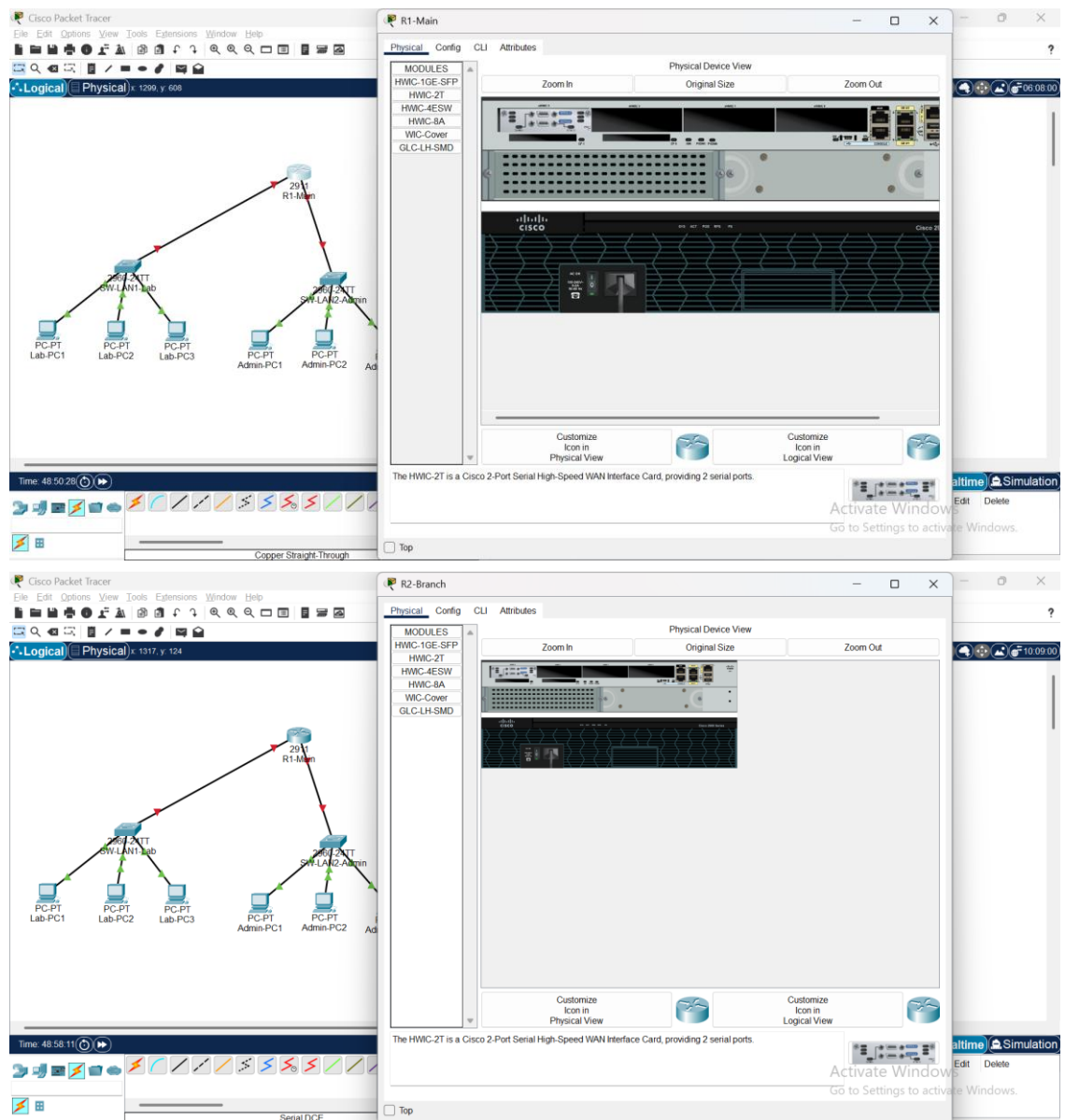
Usable IPs: 10.10.10.1 and 10.10.10.2

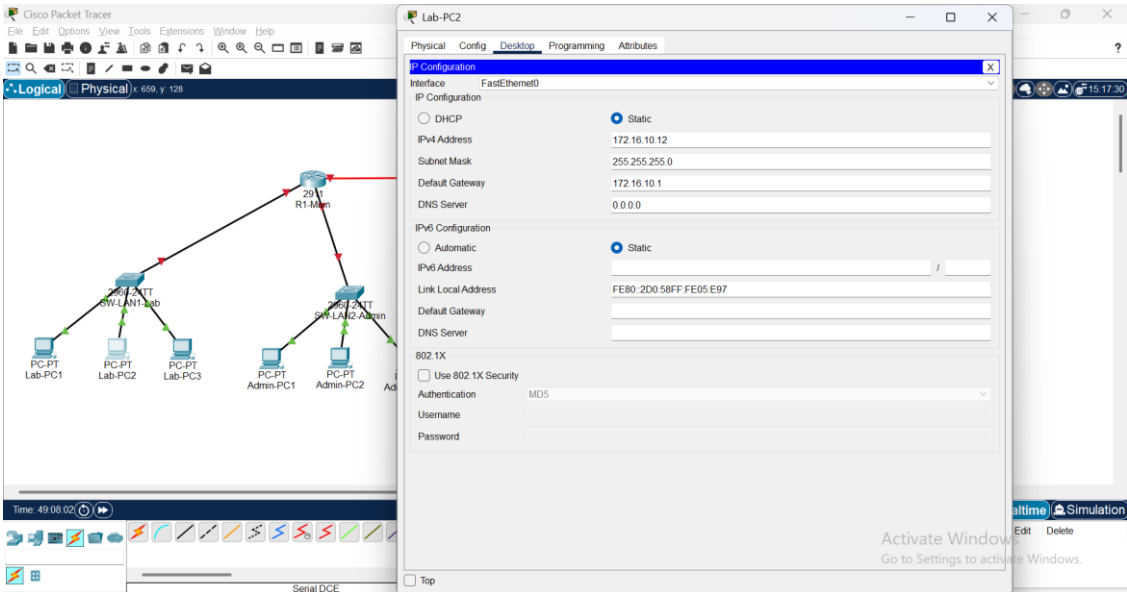
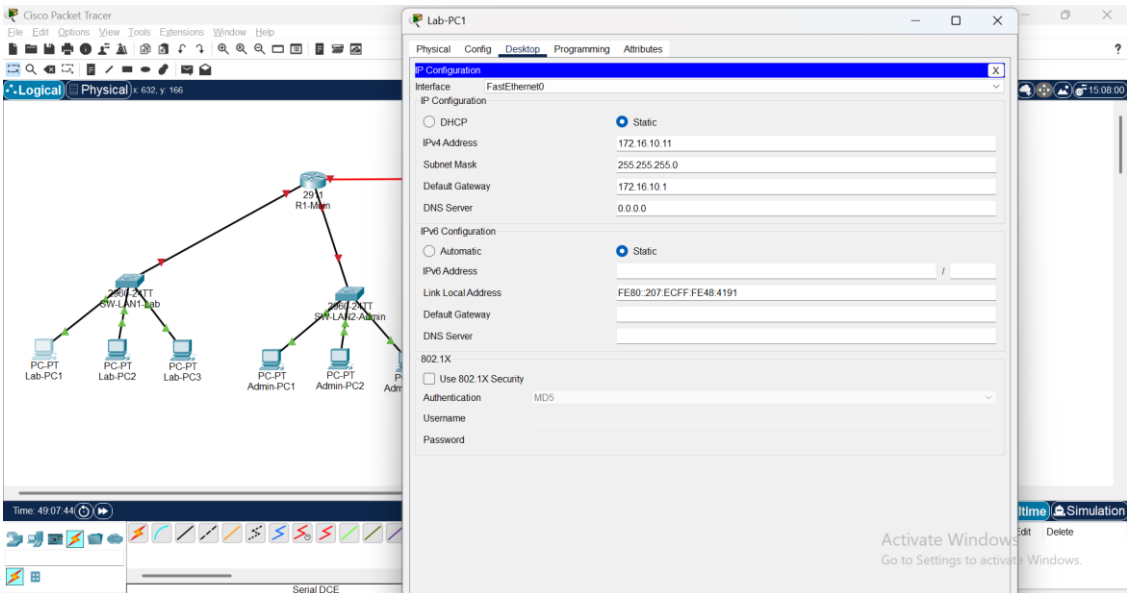
Broadcast Address: 10.10.10.3

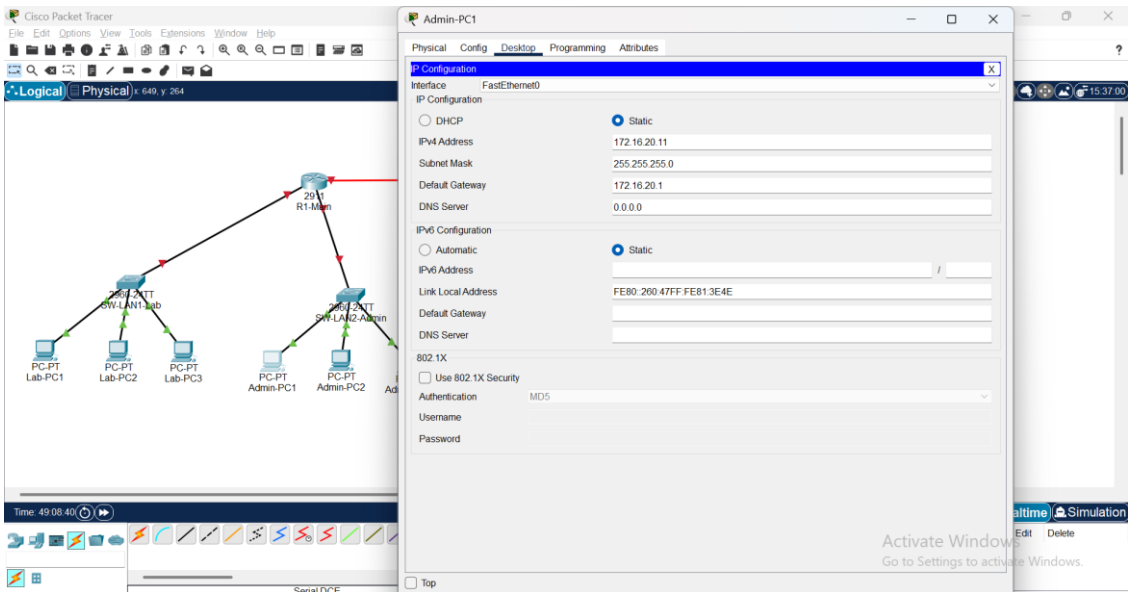
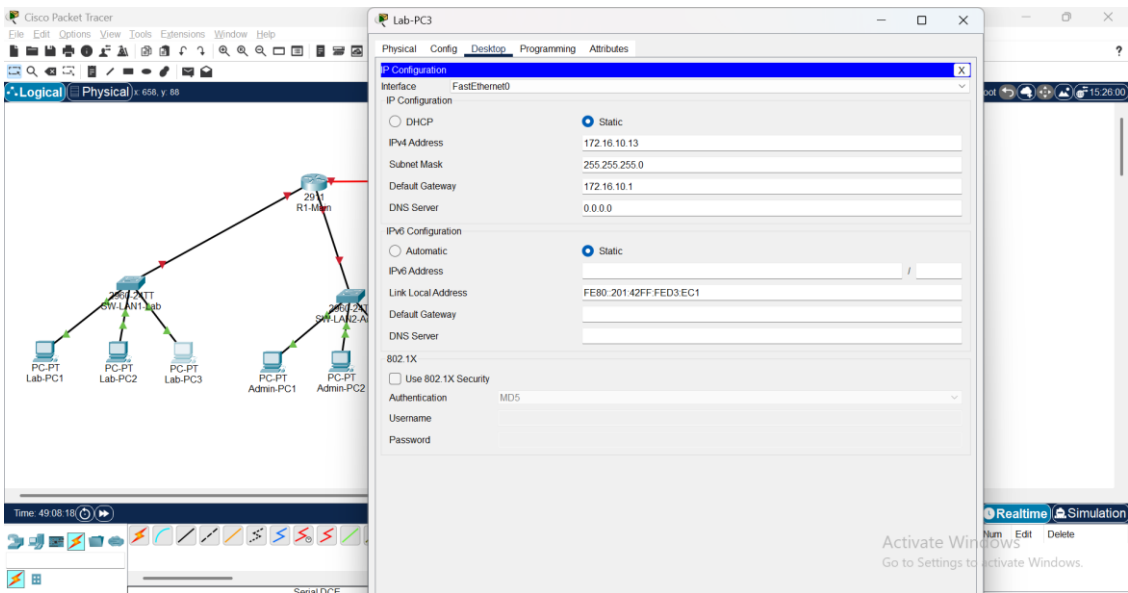


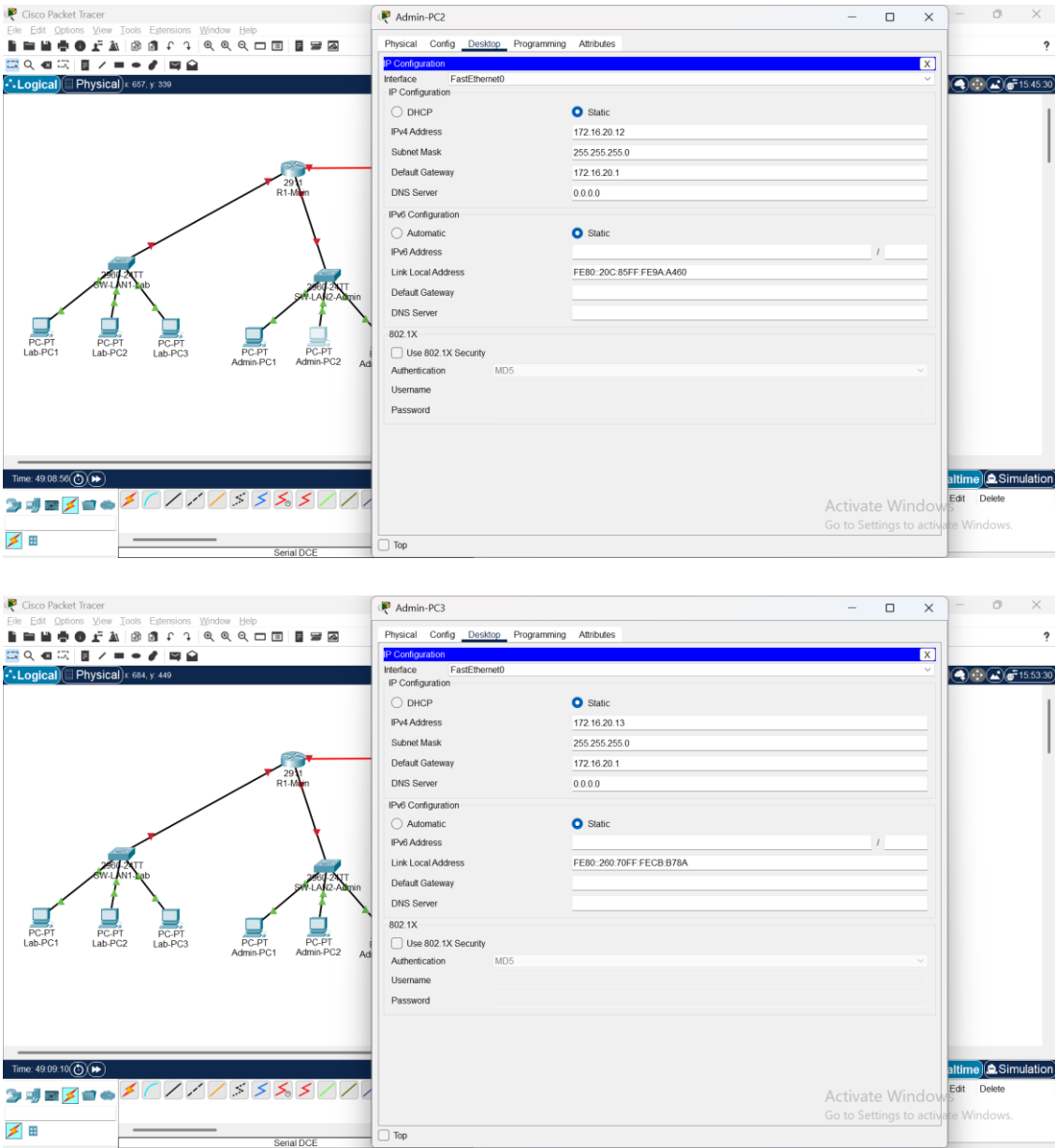


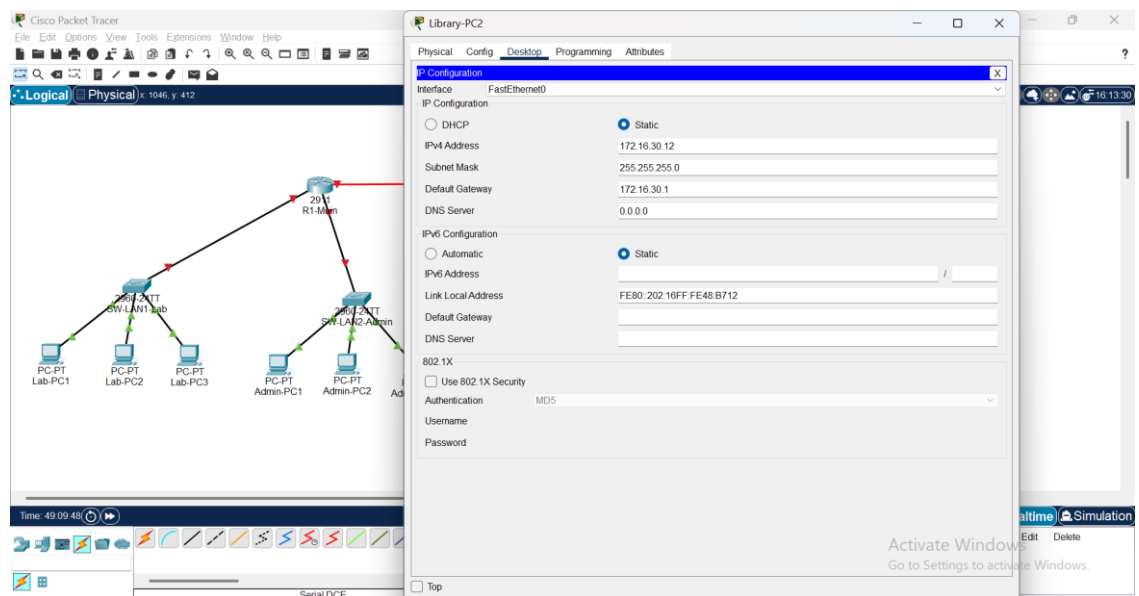
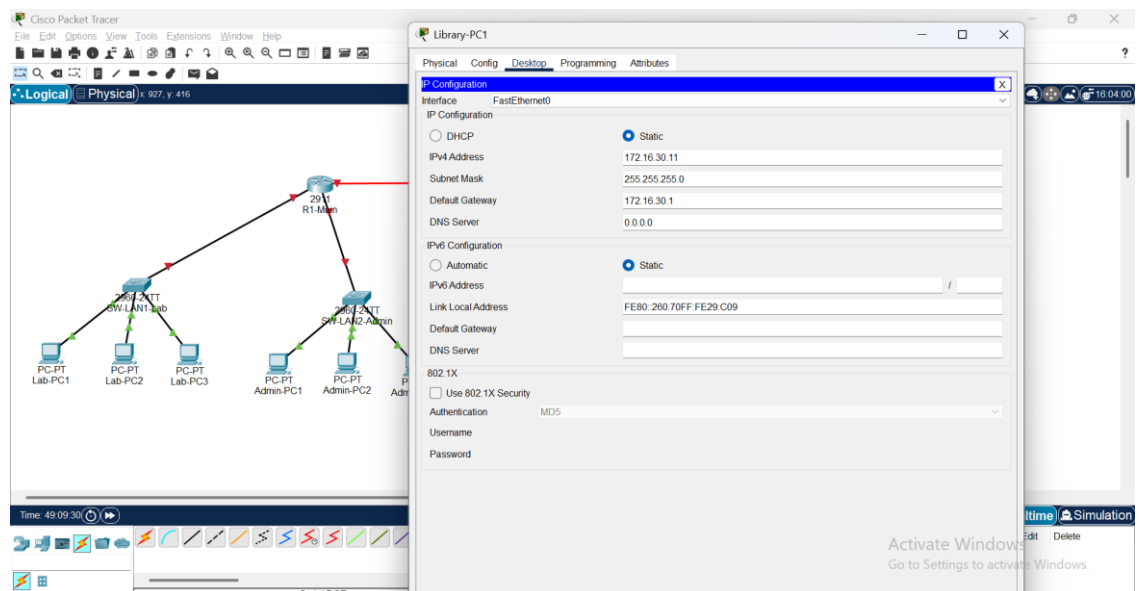


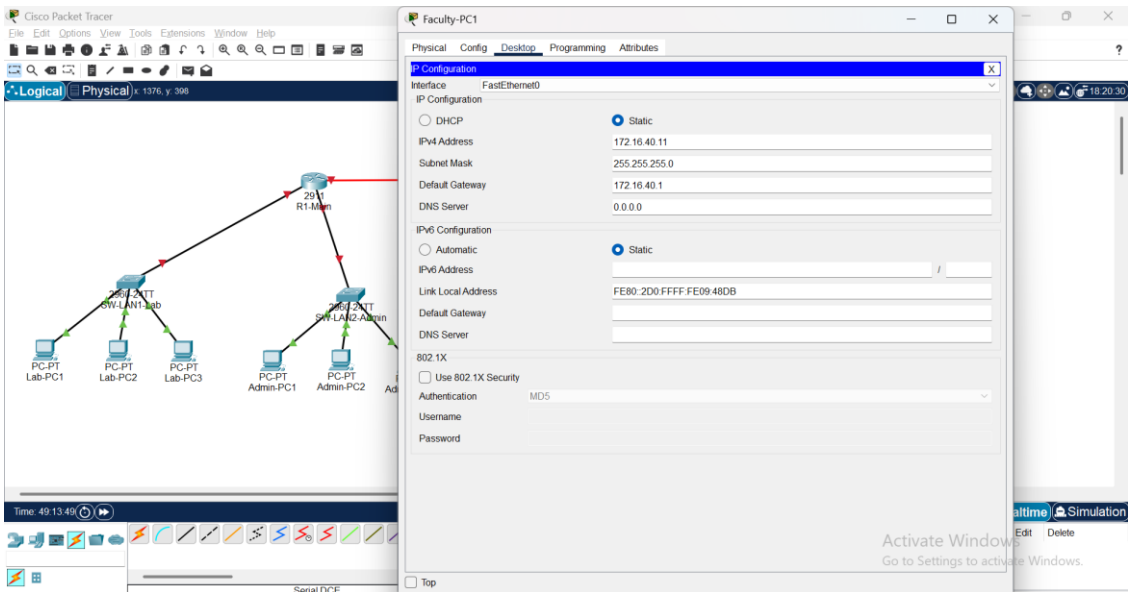
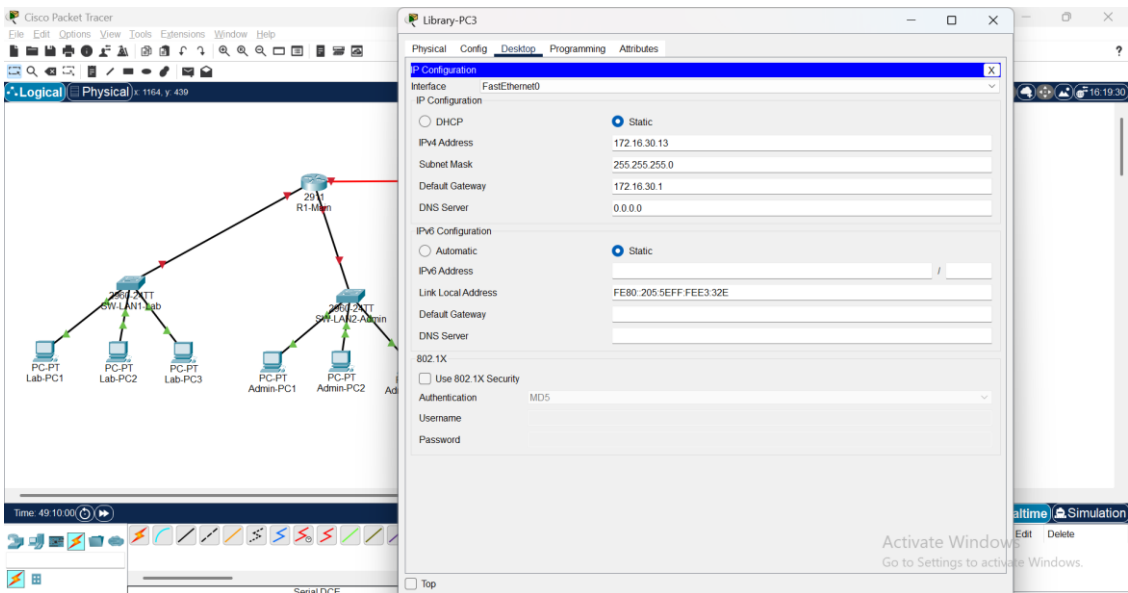


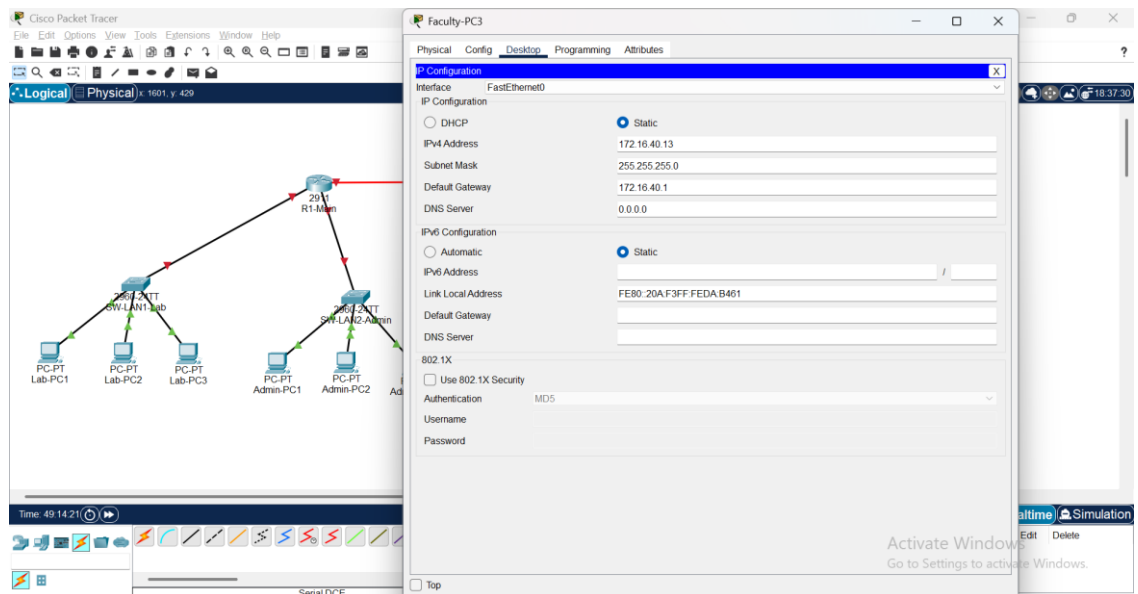
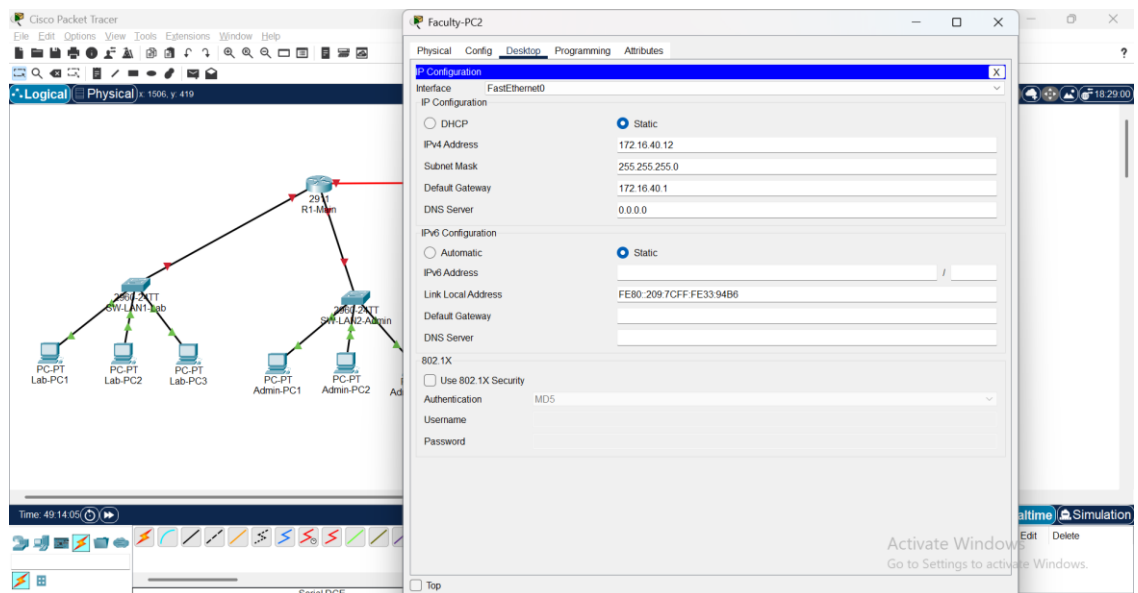


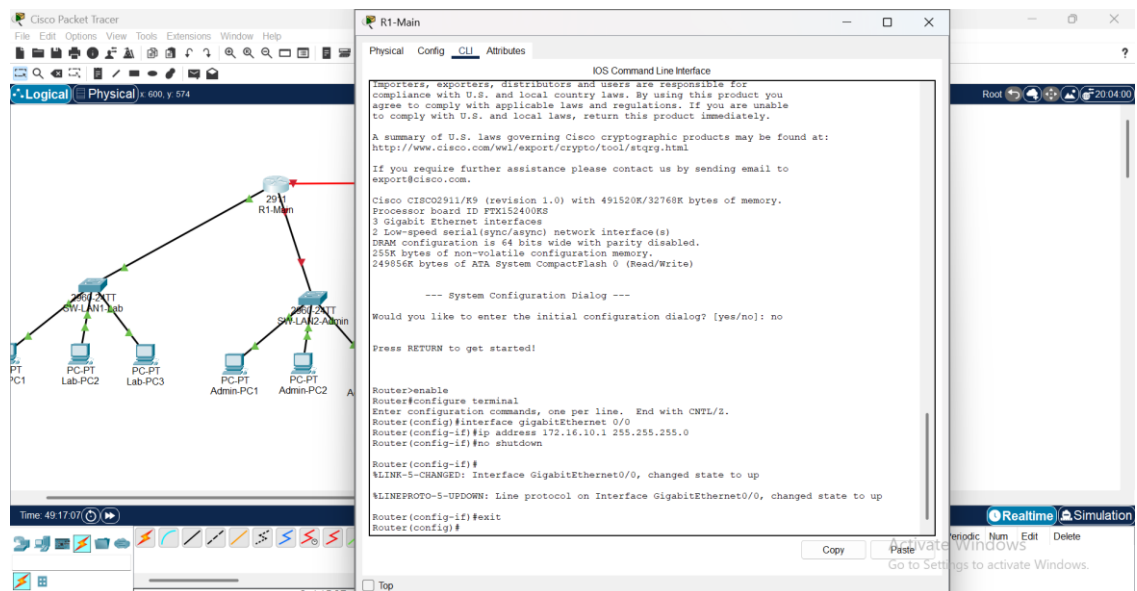












Router>enable

Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface gigabitEthernet 0/0

Router(config-if)#ip address 172.16.10.1 255.255.255.0

Router(config-if)#no shutdown

Router(config-if)#

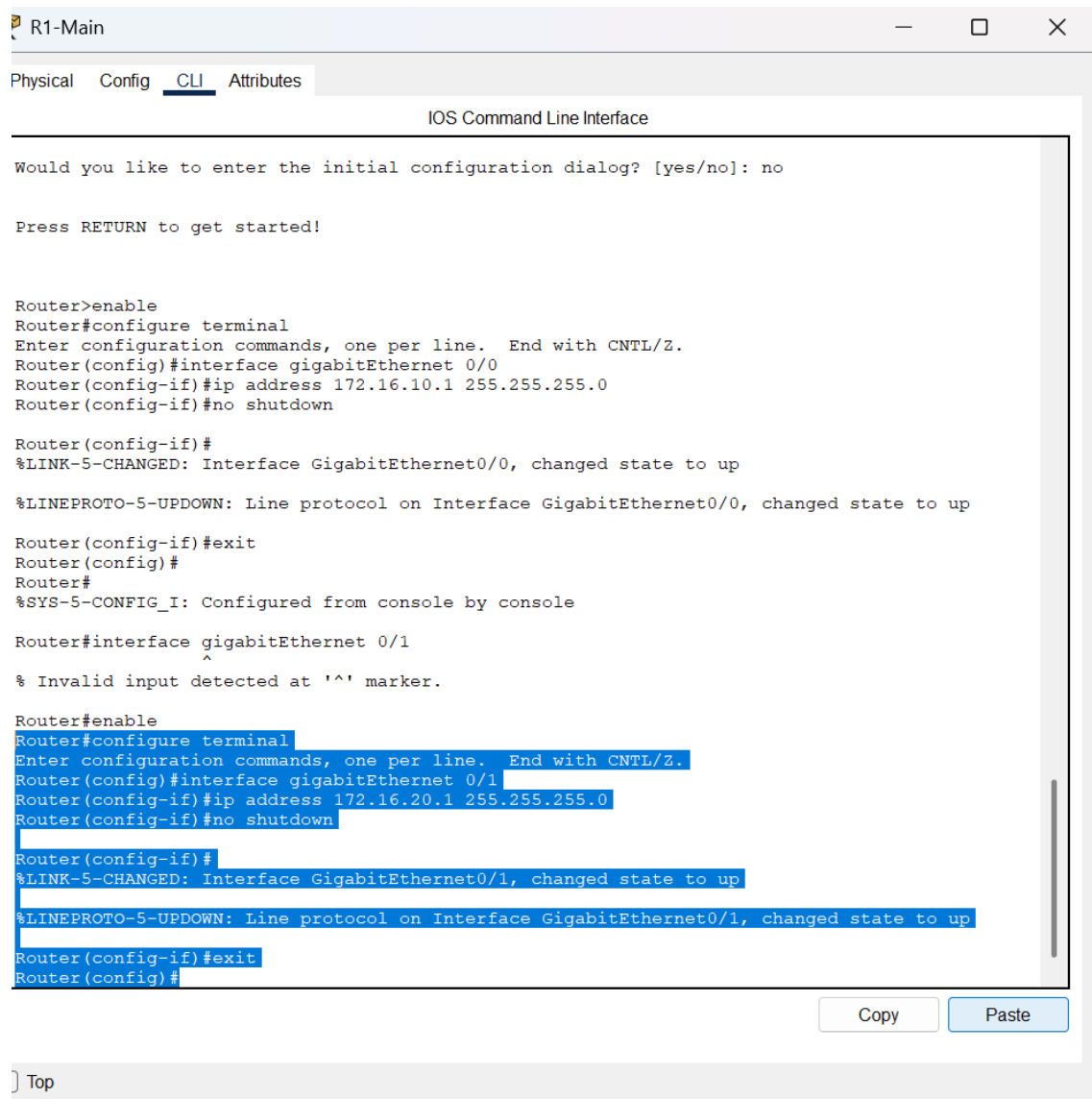
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit

Router(config)#

Router#



```
R1-Main
Physical Config CLI Attributes
IOS Command Line Interface

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 172.16.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#interface gigabitEthernet 0/1
      ^
% Invalid input detected at '^' marker.

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 172.16.20.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#
```

Copy Paste

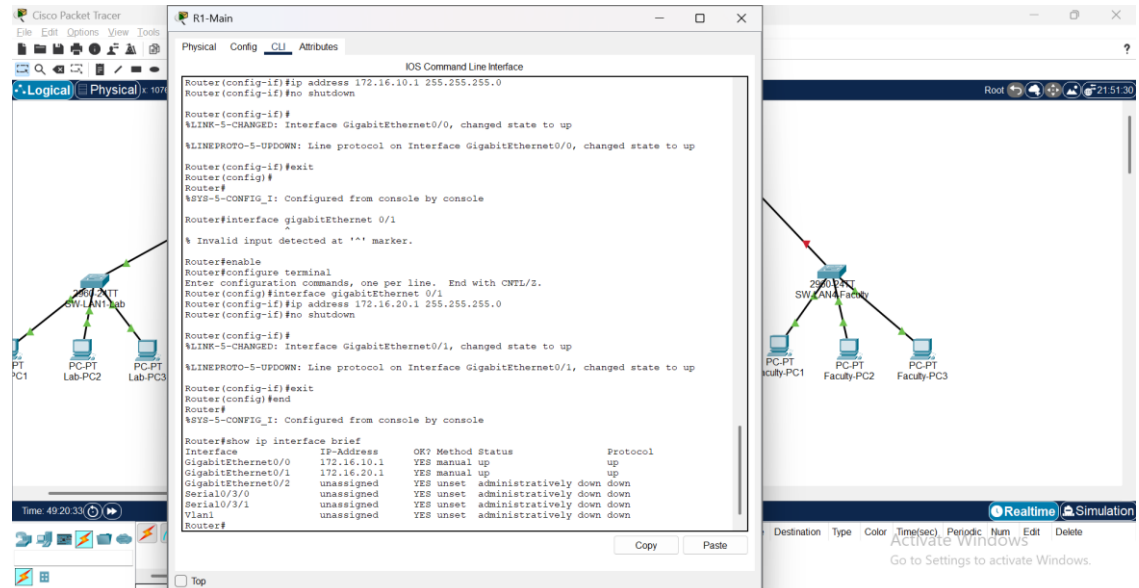
Top

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 172.16.20.1 255.255.255.0
Router(config-if)#no shutdown
```

```
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
```

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

```
Router(config-if)#exit
Router(config)#
```



```
Router(config)#end
```

```
Router#
```

```
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router#show ip interface brief
```

```
Interface IP-Address OK? Method Status Protocol
```

```
GigabitEthernet0/0 172.16.10.1 YES manual up up
```

```
GigabitEthernet0/1 172.16.20.1 YES manual up up
```

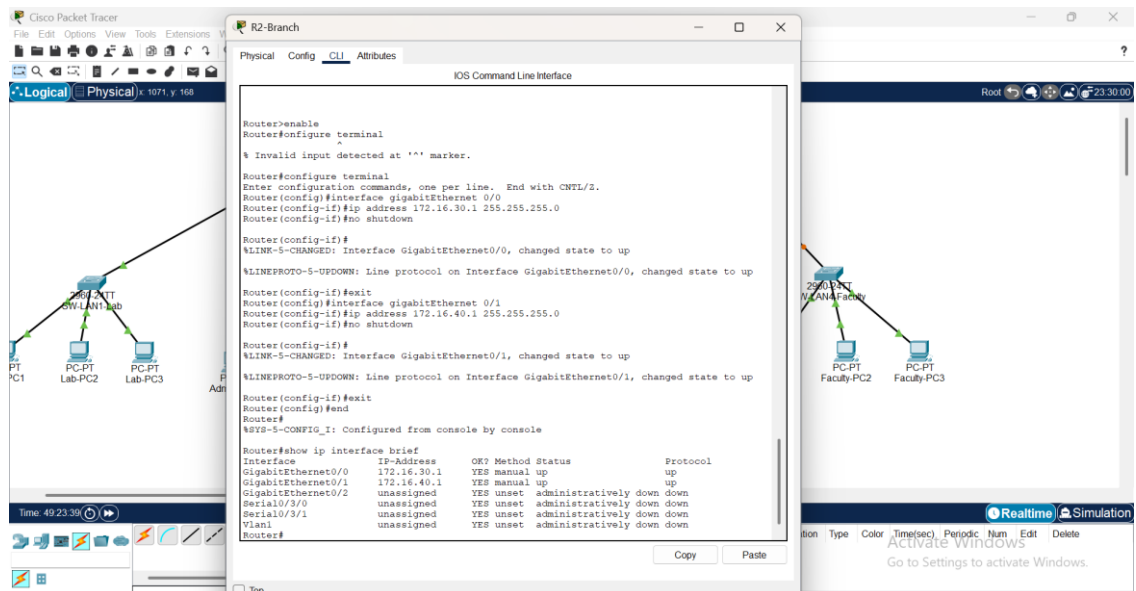
```
GigabitEthernet0/2 unassigned YES unset administratively down down
```

```
Serial0/3/0 unassigned YES unset administratively down down
```

```
Serial0/3/1 unassigned YES unset administratively down down
```

```
Vlan1 unassigned YES unset administratively down down
```

```
Router#
```



Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface gigabitEthernet 0/0

Router(config-if)#ip address 172.16.30.1 255.255.255.0

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit

Router(config)#interface gigabitEthernet 0/1

Router(config-if)#ip address 172.16.40.1 255.255.255.0

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

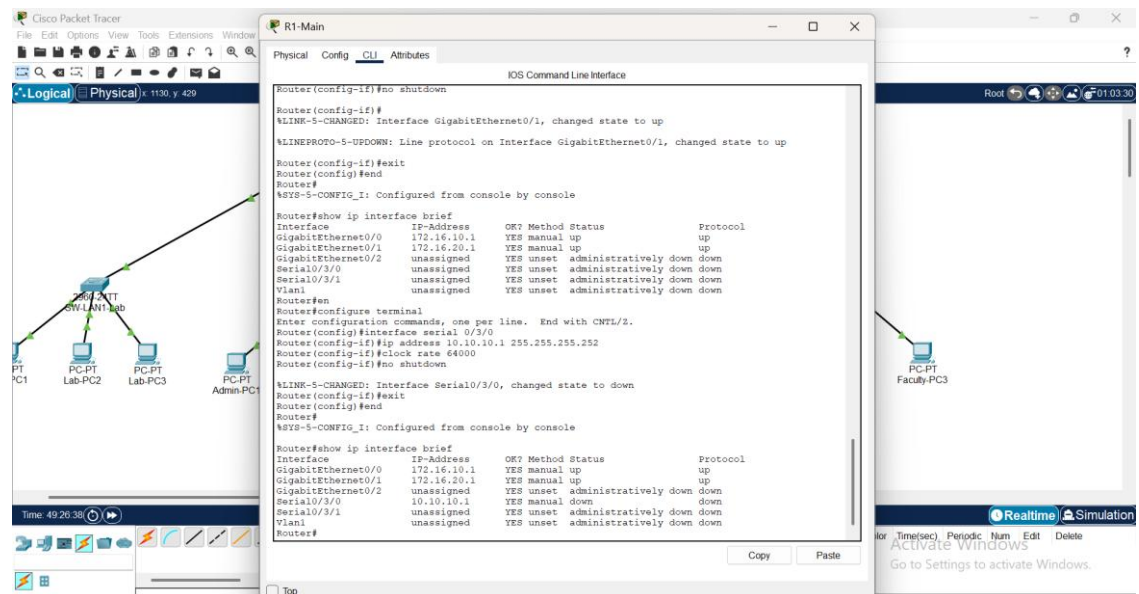
Router(config-if)#exit

Router(config)#end

Router#

%SYS-5-CONFIG_I: Configured from console by console

```
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 172.16.30.1 YES manual up up
GigabitEthernet0/1 172.16.40.1 YES manual up up
GigabitEthernet0/2 unassigned YES unset administratively down down
Serial0/3/0 unassigned YES unset administratively down down
Serial0/3/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#
```



```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface serial 0/3/0
Router(config-if)#ip address 10.10.10.1 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#no shutdown
```

```
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 172.16.10.1 YES manual up up
GigabitEthernet0/1 172.16.20.1 YES manual up up
GigabitEthernet0/2 unassigned YES unset administratively down down
Serial0/3/0 10.10.10.1 YES manual down down
Serial0/3/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#
```

R2-Branch

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 172.16.30.1 YES manual up up
GigabitEthernet0/1 172.16.40.1 YES manual up up
GigabitEthernet0/2 unassigned YES unset administratively down down
Serial0/3/0 unassigned YES unset administratively down down
Serial0/3/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface serial 0/3/0
Router(config-if)#ip address 10.10.10.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#end
Translating "end"...domain server (255.255.255.255) % Name lookup aborted
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 172.16.30.1 YES manual up up
GigabitEthernet0/1 172.16.40.1 YES manual up up
GigabitEthernet0/2 unassigned YES unset administratively down down
Serial0/3/0 10.10.10.2 YES manual up up
Serial0/3/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#
```

Copy

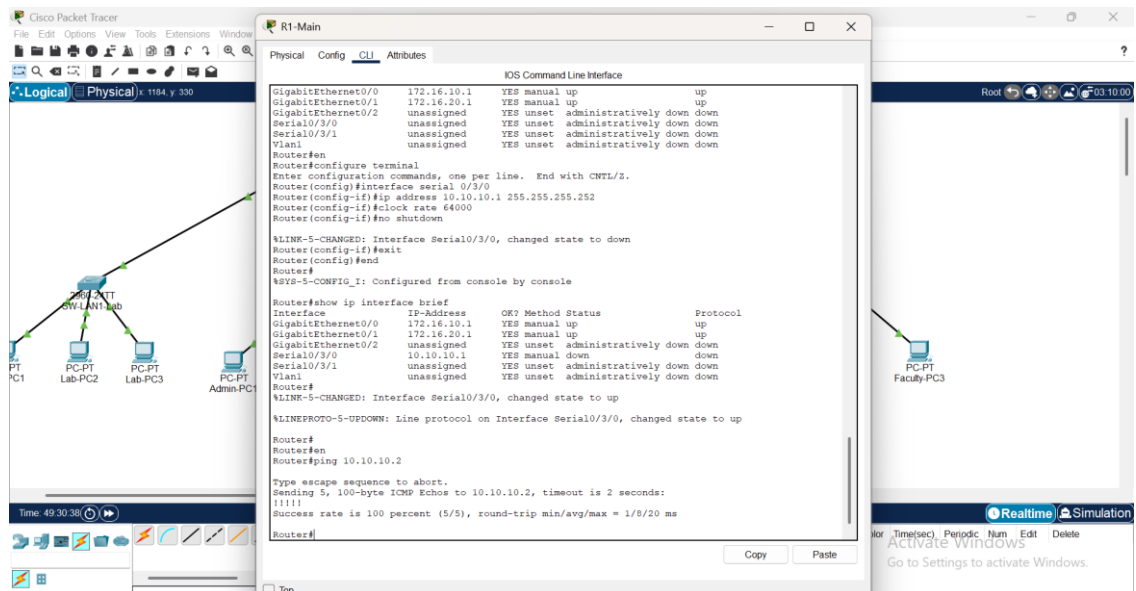
☐ Top

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface serial 0/3/0
Router(config-if)#ip address 10.10.10.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to
up
exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#end
Translating "end"...domain server (255.255.255.255) % Name lookup aborted
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 172.16.30.1 YES manual up up
GigabitEthernet0/1 172.16.40.1 YES manual up up
GigabitEthernet0/2 unassigned YES unset administratively down down
Serial0/3/0 10.10.10.2 YES manual up up
Serial0/3/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#
```



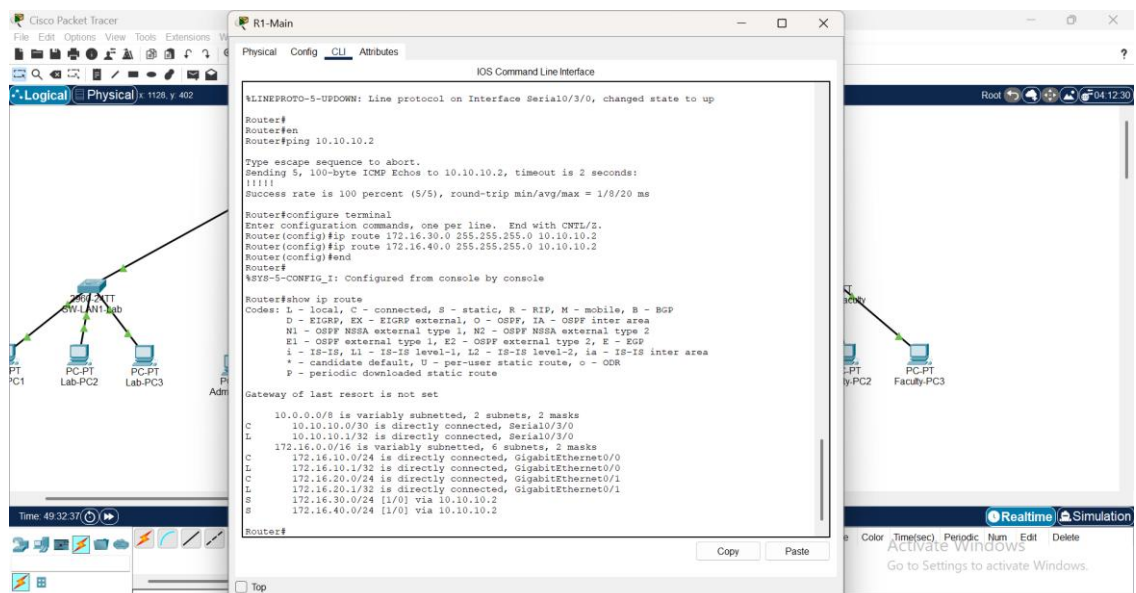
Router#ping 10.10.10.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/8/20 ms



Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

```
Router(config)#ip route 172.16.30.0 255.255.255.0 10.10.10.2
```

```
Router(config)#ip route 172.16.40.0 255.255.255.0 10.10.10.2
```

```
Router(config)#end
```

```
Router#
```

```
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router#show ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP

i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area

* - candidate default, U - per-user static route, o - ODR

P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

C 10.10.10.0/30 is directly connected, Serial0/3/0

L 10.10.10.1/32 is directly connected, Serial0/3/0

172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks

C 172.16.10.0/24 is directly connected, GigabitEthernet0/0

L 172.16.10.1/32 is directly connected, GigabitEthernet0/0

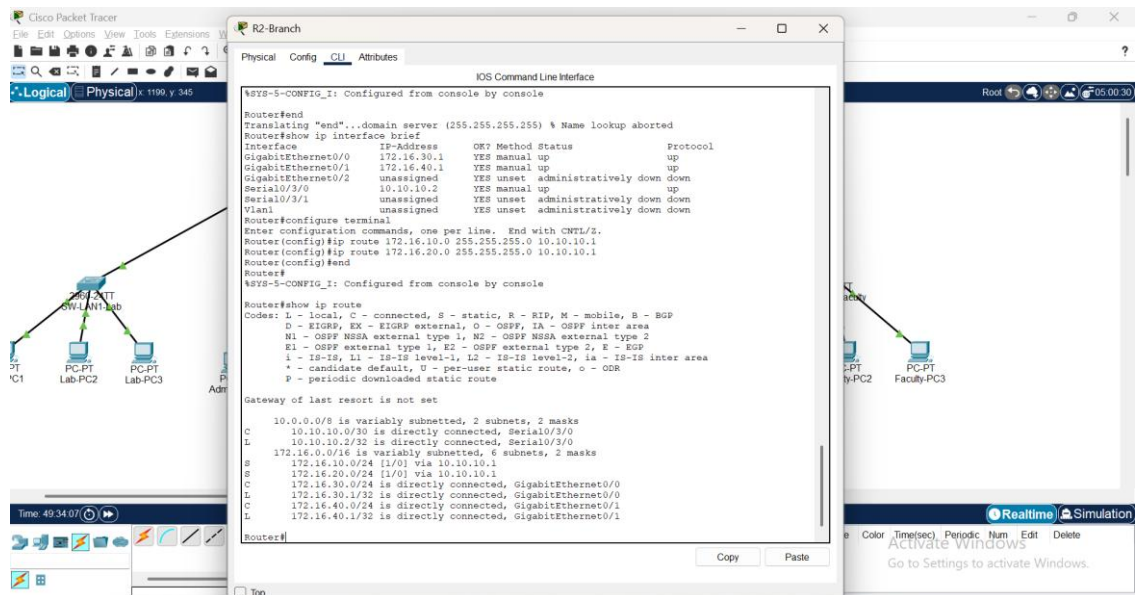
C 172.16.20.0/24 is directly connected, GigabitEthernet0/1

L 172.16.20.1/32 is directly connected, GigabitEthernet0/1

S 172.16.30.0/24 [1/0] via 10.10.10.2

S 172.16.40.0/24 [1/0] via 10.10.10.2

```
Router#
```



Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#ip route 172.16.10.0 255.255.255.0 10.10.10.1

Router(config)#ip route 172.16.20.0 255.255.255.0 10.10.10.1

Router(config)#end

Router#

%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP

i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area

* - candidate default, U - per-user static route, o - ODR

P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

C 10.10.10.0/30 is directly connected, Serial0/3/0

L 10.10.10.2/32 is directly connected, Serial0/3/0

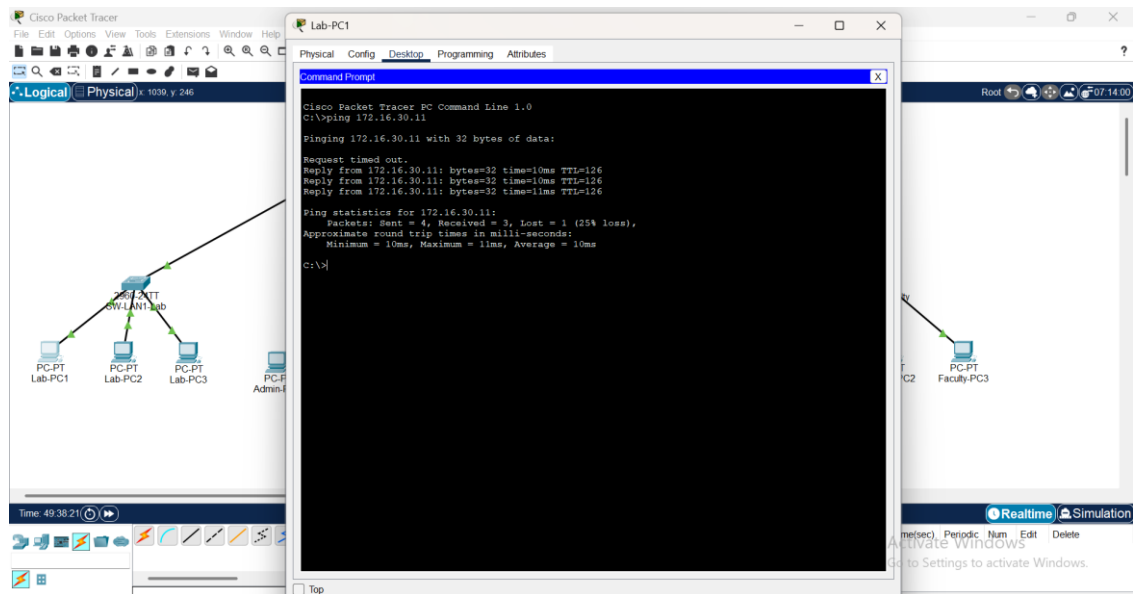
172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks

S 172.16.10.0/24 [1/0] via 10.10.10.1

S 172.16.20.0/24 [1/0] via 10.10.10.1

C 172.16.30.0/24 is directly connected, GigabitEthernet0/0
L 172.16.30.1/32 is directly connected, GigabitEthernet0/0
C 172.16.40.0/24 is directly connected, GigabitEthernet0/1
L 172.16.40.1/32 is directly connected, GigabitEthernet0/1

Router#



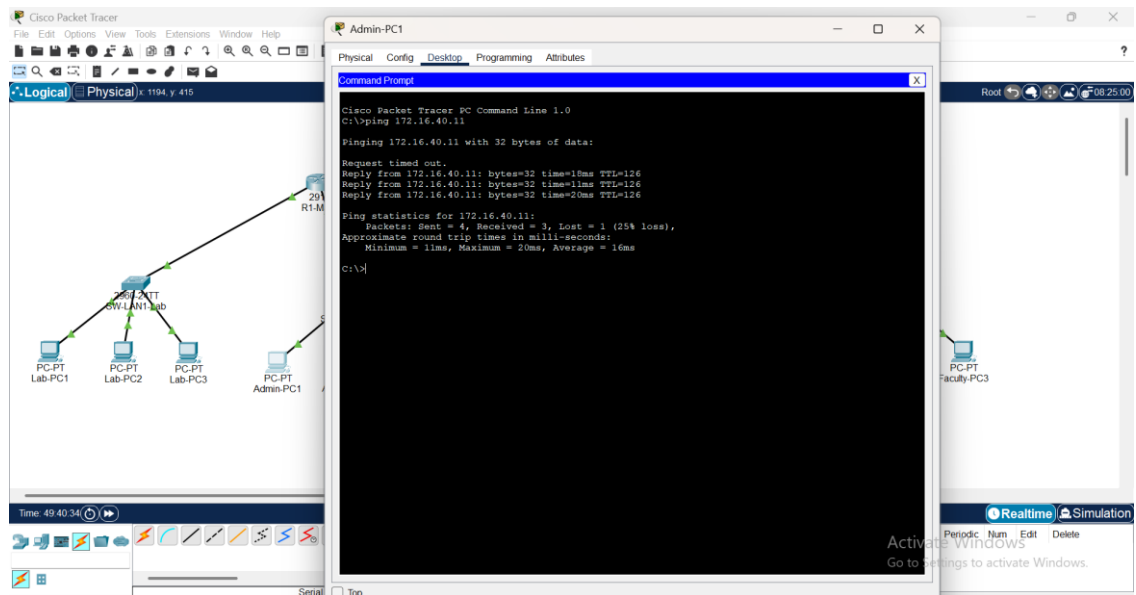
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.30.11

Pinging 172.16.30.11 with 32 bytes of data:

Request timed out.
Reply from 172.16.30.11: bytes=32 time=10ms TTL=126
Reply from 172.16.30.11: bytes=32 time=10ms TTL=126
Reply from 172.16.30.11: bytes=32 time=11ms TTL=126

Ping statistics for 172.16.30.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 11ms, Average = 10ms

C:\>
C:\>
```



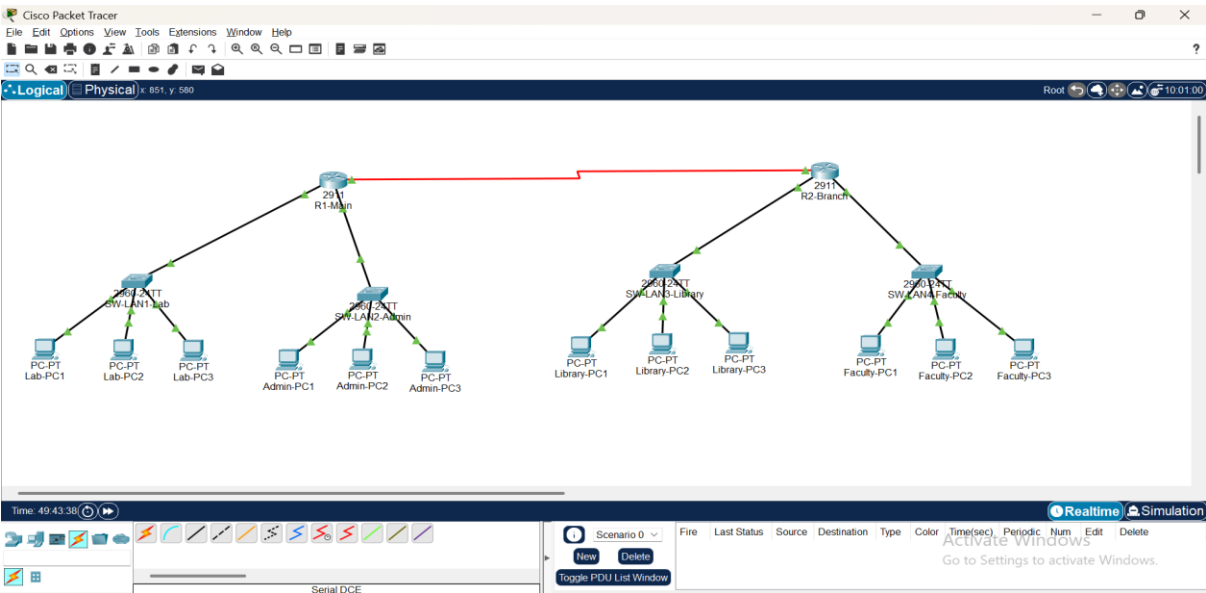
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.40.11

Pinging 172.16.40.11 with 32 bytes of data:

Request timed out.
Reply from 172.16.40.11: bytes=32 time=18ms TTL=126
Reply from 172.16.40.11: bytes=32 time=11ms TTL=126
Reply from 172.16.40.11: bytes=32 time=20ms TTL=126

Ping statistics for 172.16.40.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 20ms, Average = 16ms

C:\>
```



- Reflection :-

One Key Decision Made by the Team: Our team decided to use static routing because it is easy to configure and suitable for the network required in this phase. One Problem Encountered and How It Was Solved: We had a problem with the IP address configuration, which caused the devices to fail to communicate. After checking the IP addresses and router settings, the issue was resolved successfully. Plan for the Next Reporting Period: In the next phase, we will continue testing the network, verify all configurations, and complete the remaining project requirements

Project Artifacts:-

1- Word Online.

https://seuedu-my.sharepoint.com/:w:/g/personal/s220041205_seu_edu_sa/IQAIA8uF0tEHSa0Hou8a50CDAc0WbP54oROG4W_NqugOFGc

2- GitHub Repository.

<https://github.com/Dalal-ali-ops/IT351-Project.git>

(15 Marks)**Phase 2:- Phase 2 Requirements****Testing and Verification**

1. Use the ping command to test connectivity between a device in any LAN of WAN 1 and a device in any LAN of WAN 2.
2. Verify that data packets are successfully transmitted (0% packet loss).

Questions to Answer (10 Marks — 2 marks each)

Answer all five questions. For each question, you must include (a) a clear screenshot from your Packet Tracer project and (b) the commands used at that instance (e.g., IOS CLI commands or the configuration steps applied). Screenshots without the corresponding commands/steps will receive ZERO marks for that question.

1. **Creating nodes:** Show a screenshot of the complete topology with all nodes created (12 end devices, 4 switches, 2 routers). List the device types selected and any steps/commands used to name the devices (e.g., hostname configuration). (2 marks)
2. **Giving connections in the LAN:** Show a screenshot of the LAN connections between end devices, switches, and routers. State the cable types used for each connection and why. (2 marks)
3. **Assigning IP addresses:** Show screenshots of IP address, subnet mask, and default gateway assignment on at least one end device in each LAN, and the commands used to configure the router LAN interfaces (e.g., interface, ip address, no shutdown). (2 marks)
4. **Creating the WAN connection:** Show a screenshot of the serial link between Router 1 and Router 2, and the commands used to configure the serial interfaces and enable routing (static routes or a dynamic routing protocol such as RIP), including clock rate if applicable. (2 marks)
5. **Successful data transmission using the ping command:** Show screenshots of successful ping tests between a device in WAN 1 and a device in WAN 2 (both directions), with the exact ping commands used and the results (replies received, 0% loss). (2 marks)

Answer:

IP Addressing Scheme

1. Creating Nodes

After completing the initial network design in Phase 1, we continued by creating the required devices in Cisco Packet Tracer. The topology includes 12 end devices, four switches, and two routers, divided between the Main Campus and Branch Campus. The Main Campus contains the Computer Lab and Admin Office, while the Branch Campus contains the Library and Faculty Office. Each LAN includes three end devices connected to an access switch

The following device types were used:

- 12 PC-PT end devices
- 4 Cisco 2960 switches
- 2 Cisco routers
- Serial interface modules, where required

To make the devices easy to identify during configuration and testing, they were renamed according to their campus and department. The PCs were named **Lab-PC1 to Lab-PC3**, **Admin-PC1 to Admin-PC3**, **Library-PC1 to Library-PC3**, and **Faculty-PC1 to Faculty-PC3**. The routers were named **R1-Main** and **R2-Branch**.

The device names were configured through the **Config** tab in Packet Tracer by entering the required name in the **Display Name** field.

```
Router> enable
```

```
Router# configure terminal
```

```
Router(config)# hostname R1-Main
```

```
R1-Main(config)# end
```

LAN 1 – Computer Lab

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1-Main	GigabitEthernet0/0	172.16.10.1	255.255.255.0	—
Lab-PC1	FastEthernet0	172.16.10.11	255.255.255.0	172.16.10.1
Lab-PC2	FastEthernet0	172.16.10.12	255.255.255.0	172.16.10.1
Lab-PC3	FastEthernet0	172.16.10.13	255.255.255.0	172.16.10.1

Network Address: 172.16.10.0/24

Broadcast Address: 172.16.10.255

Router> enable

Router# configure terminal

Router(config)# hostname R2-Branch

R2-Branch(config)# end

LAN 2 – Admin Office

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1-Main	GigabitEthernet0/1	172.16.20.1	255.255.255.0	—
Admin-PC1	FastEthernet0	172.16.20.11	255.255.255.0	172.16.20.1
Admin-PC2	FastEthernet0	172.16.20.12	255.255.255.0	172.16.20.1
Admin-PC3	FastEthernet0	172.16.20.13	255.255.255.0	172.16.20.1

Network Address: 172.16.20.0/24

Broadcast Address: 172.16.20.255

LAN 3 – Library

Device	Interface	IP Address	Subnet Mask	Default Gateway
R2-Branch	GigabitEthernet0/0	172.16.30.1	255.255.255.0	—
Library-PC1	FastEthernet0	172.16.30.11	255.255.255.0	172.16.30.1
Library-PC2	FastEthernet0	172.16.30.12	255.255.255.0	172.16.30.1
Library-PC3	FastEthernet0	172.16.30.13	255.255.255.0	172.16.30.1

Network Address: 172.16.30.0/24

Broadcast Address: 172.16.30.255

LAN 4 – Faculty Office

Device	Interface	IP Address	Subnet Mask	Default Gateway
R2-Branch	GigabitEthernet0/1	172.16.40.1	255.255.255.0	—
Faculty-PC1	FastEthernet0	172.16.40.11	255.255.255.0	172.16.40.1
Faculty-PC2	FastEthernet0	172.16.40.12	255.255.255.0	172.16.40.1
Faculty-PC3	FastEthernet0	172.16.40.13	255.255.255.0	172.16.40.1

Network Address: 172.16.40.0/24

Broadcast Address: 172.16.40.255

WAN Serial Link

Device	Interface	IP Address	Subnet Mask
R1-Main	Serial0/3/0	10.10.10.1	255.255.255.252
R2-Branch	Serial0/3/0	10.10.10.2	255.255.255.252

Network Address: 10.10.10.0/30

Usable IPs: 10.10.10.1 and 10.10.10.2

Broadcast Address: 10.10.10.3

2. Giving connections in the LAN

Each end device was connected to its local switch using a **copper straight-through cable**, which is suitable for connecting different device types such as PCs and switches. Similarly, each switch was connected to its corresponding router through the **Gigabit Ethernet interface** using a copper straight-through cable. The two routers were connected using a **serial DCE/DTE cable** to establish the WAN connection between the Main Campus and Branch Campus. The **DCE side** provides the required clocking for the serial connection, so the **clock rate** was configured on the router connected to the DCE end.

Connection	Cable Type	Reason
PC to Switch	Copper Straight-Through	Used to connect a PC to a switch
Switch to Router	Copper Straight-Through	Used to connect a switch to a router
Router 1 to Router 2	Serial DCE/DTE	Used to establish the WAN connection
DCE serial interface	Serial DCE with clock rate	Provides clocking for the serial connection

3. Assigning IP addresses

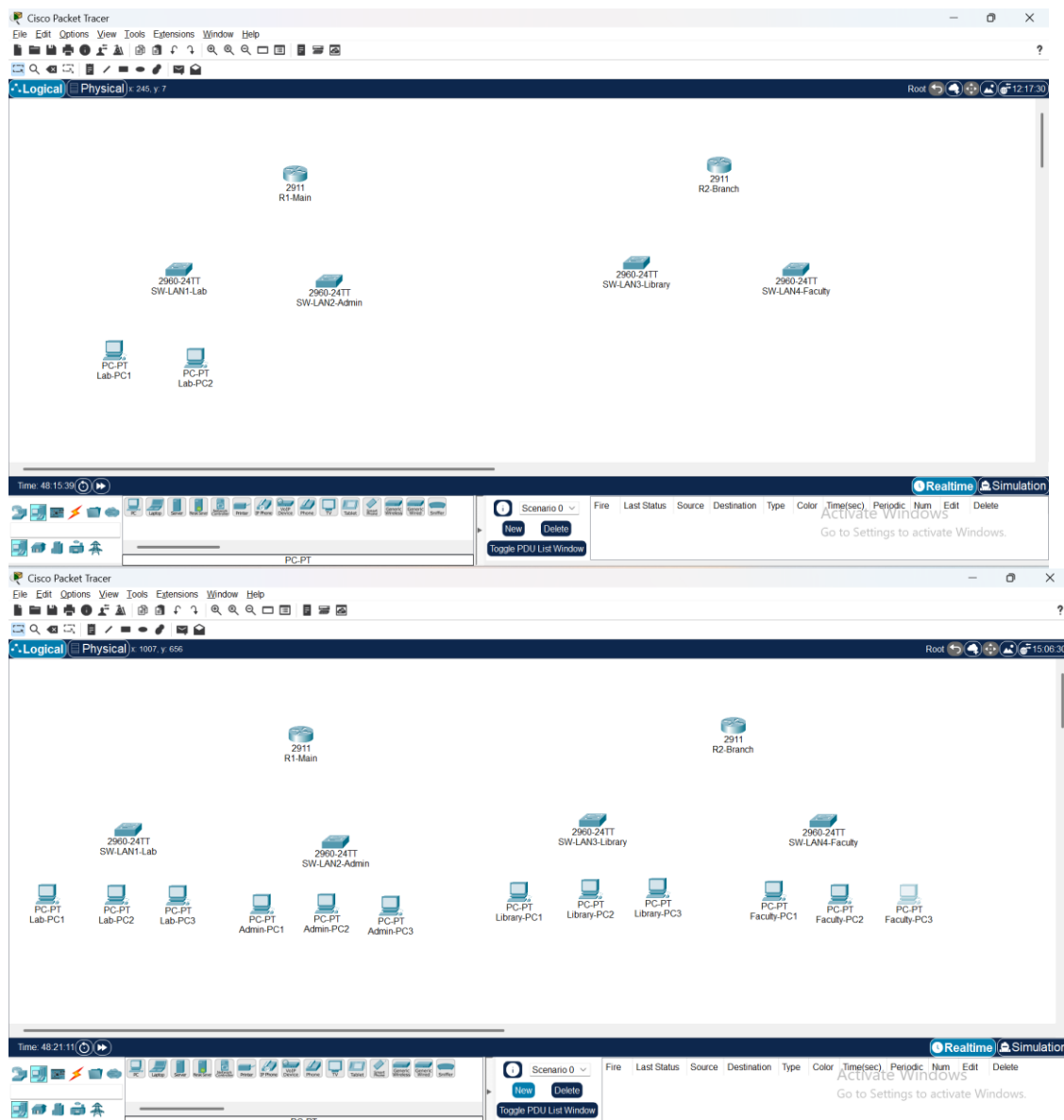
The following screenshots show the IP addressing configuration for the end devices and router interfaces.

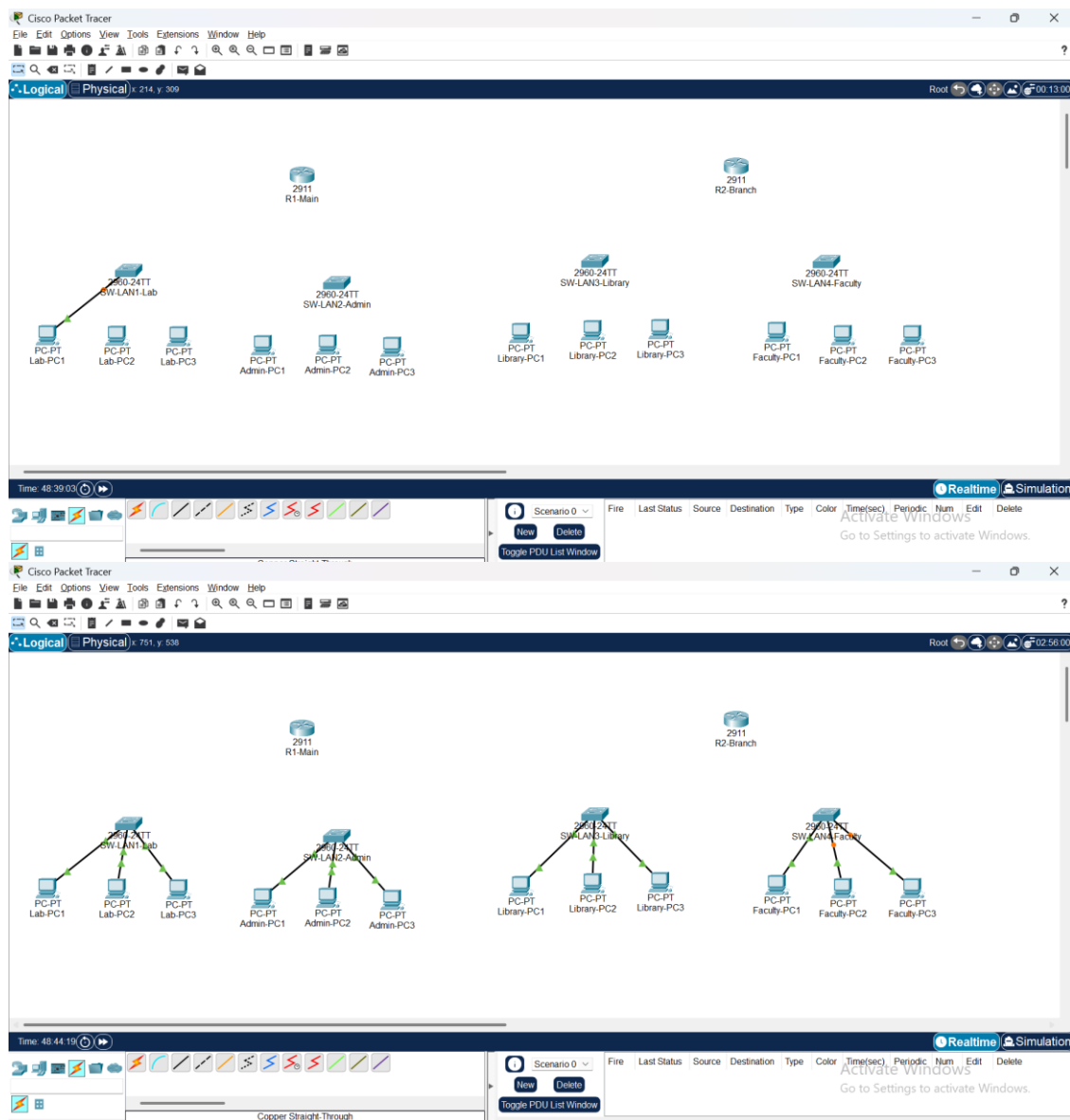
4. Creating WAN connection

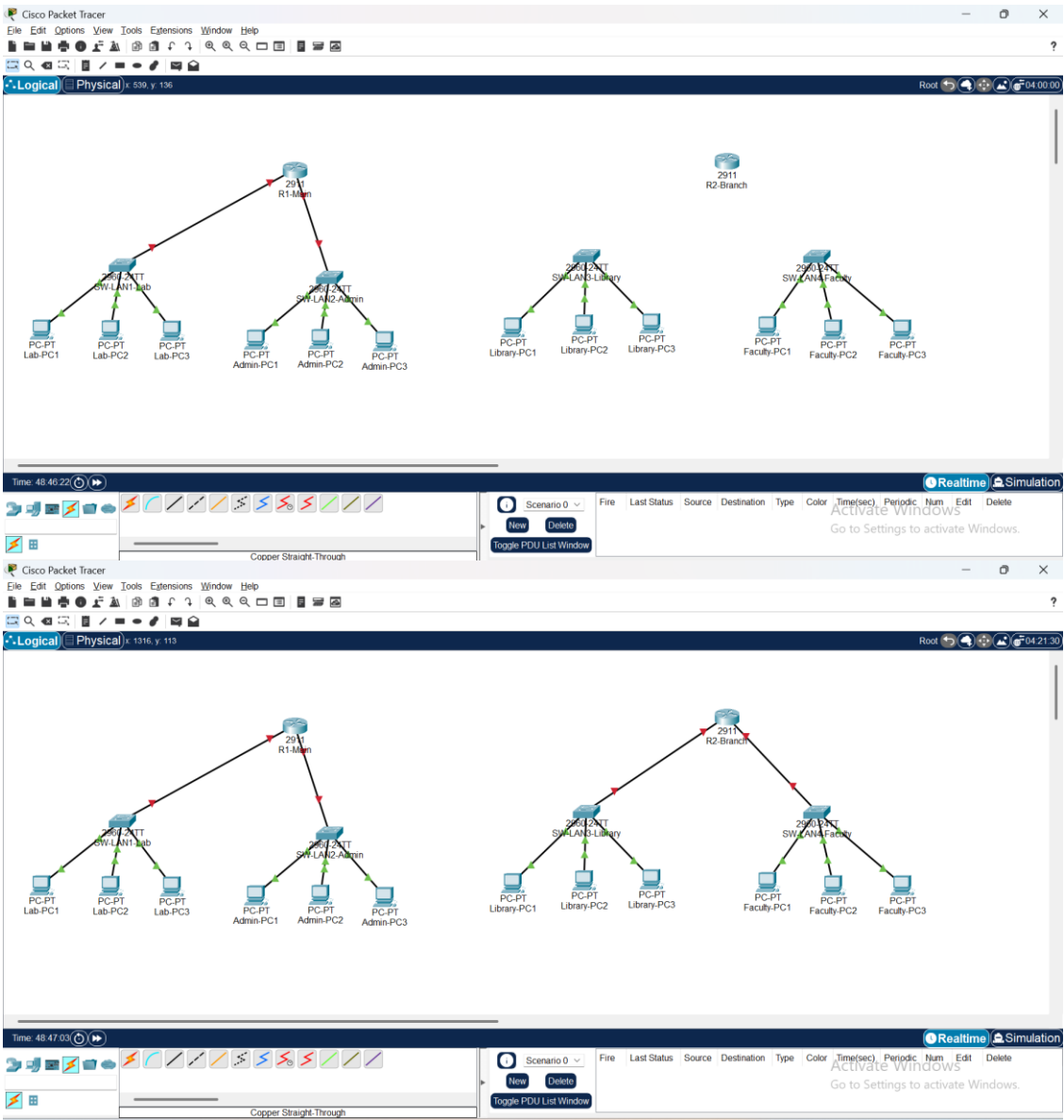
The following screenshots show the configuration of the WAN connection between R1-Main and R2-Branch, including the serial interface and routing configuration.

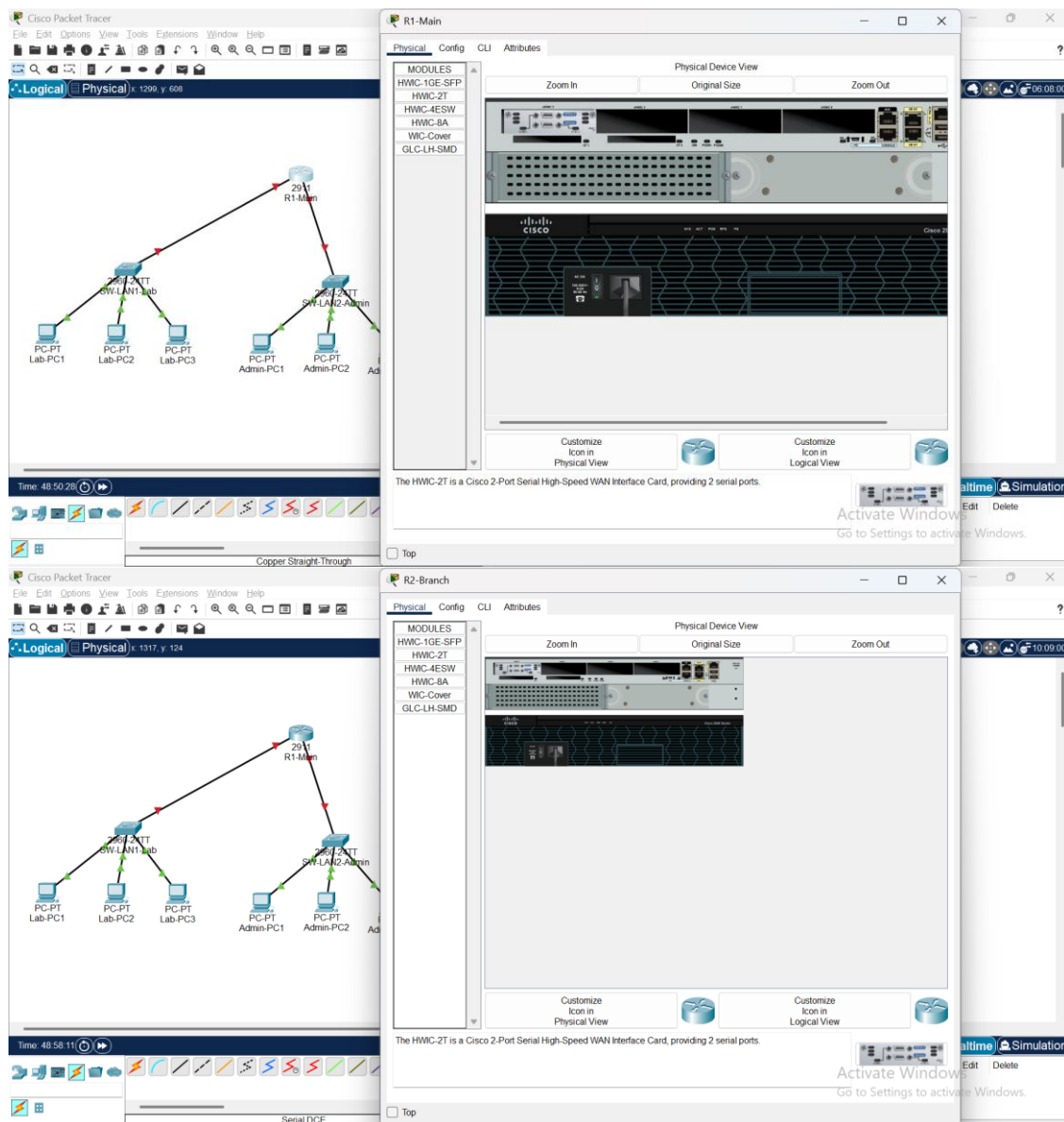
5. Successful data transmission

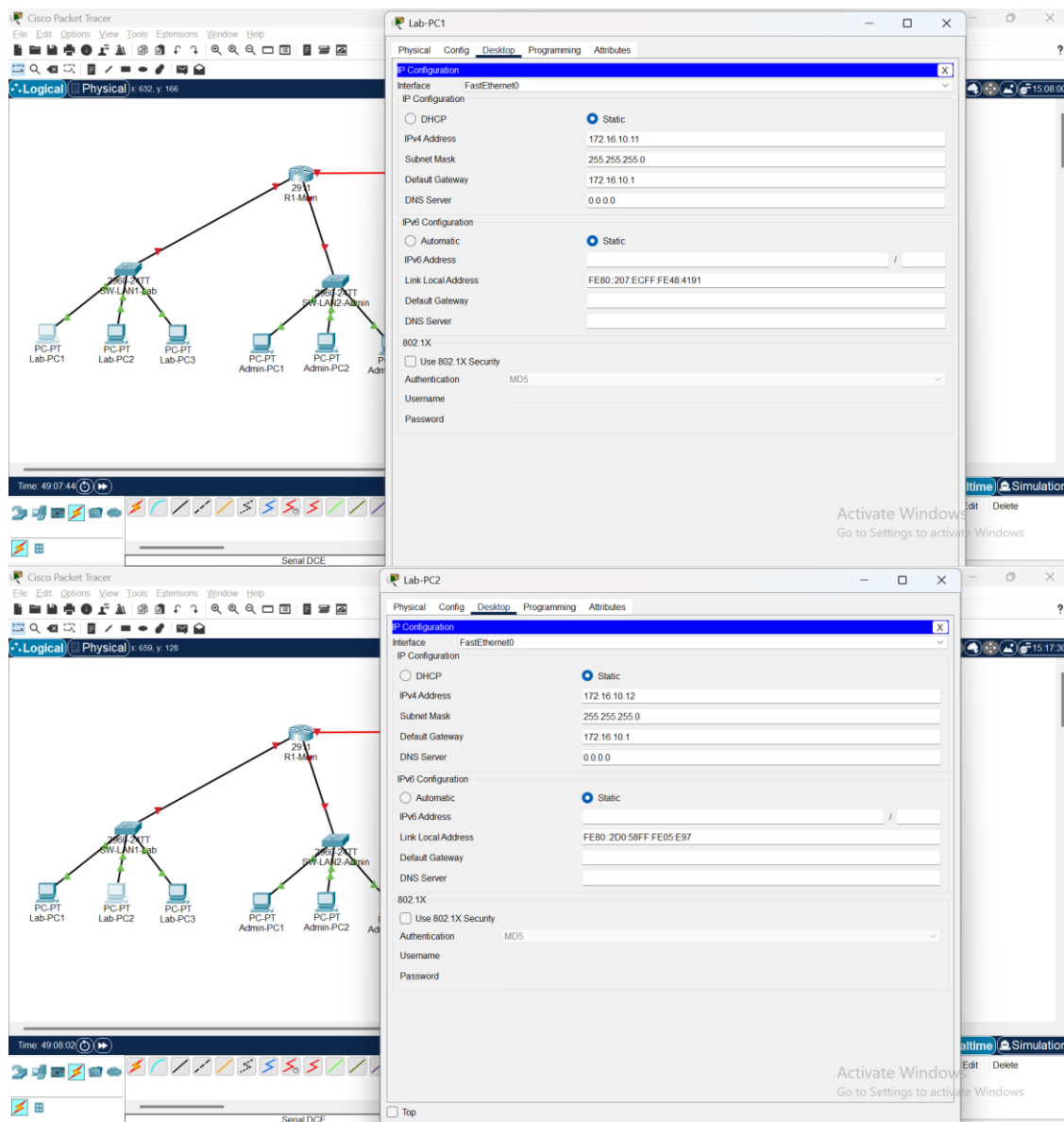
The following screenshots show the connectivity tests performed between devices in the different WANs to verify successful data transmission.

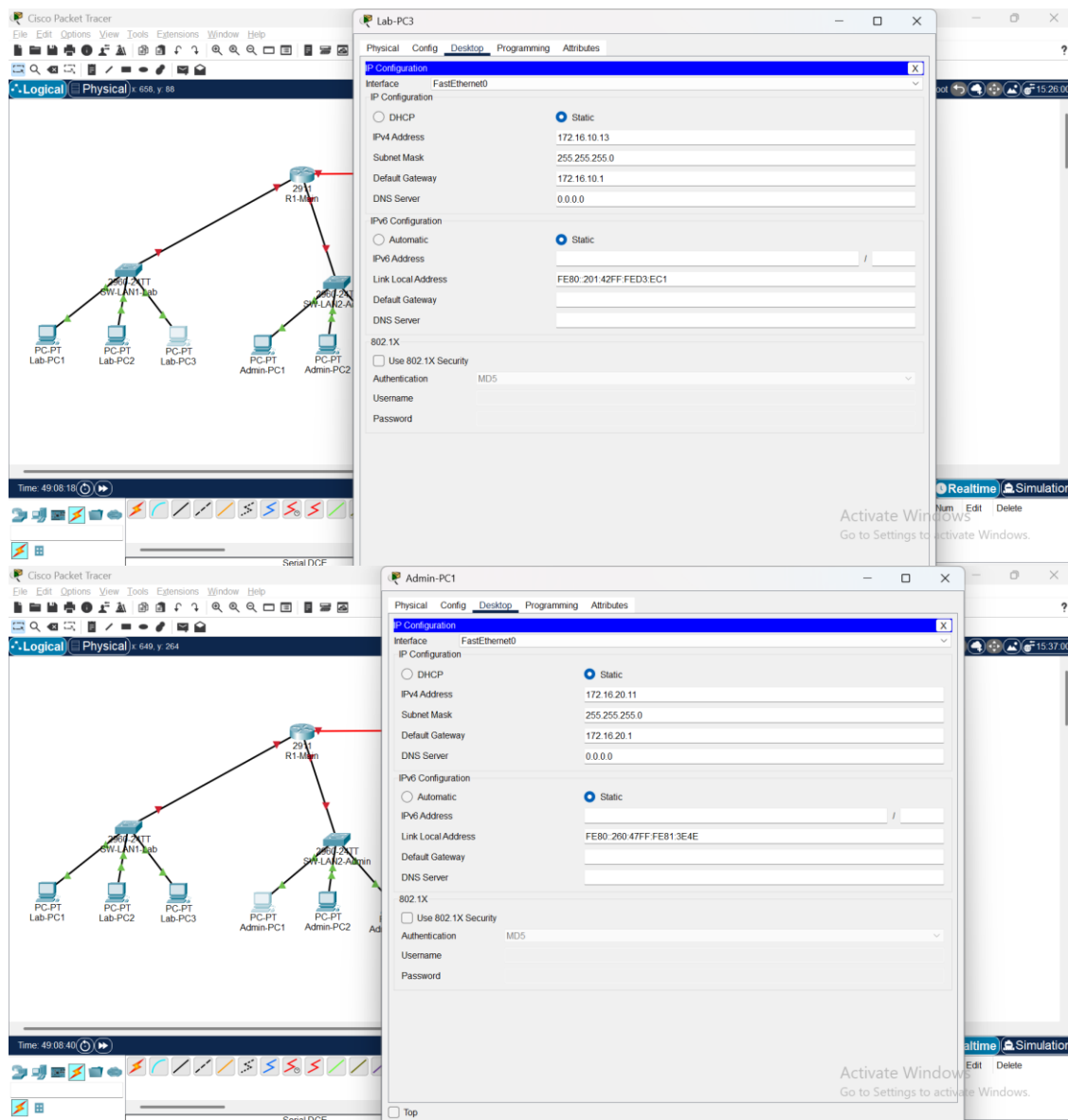


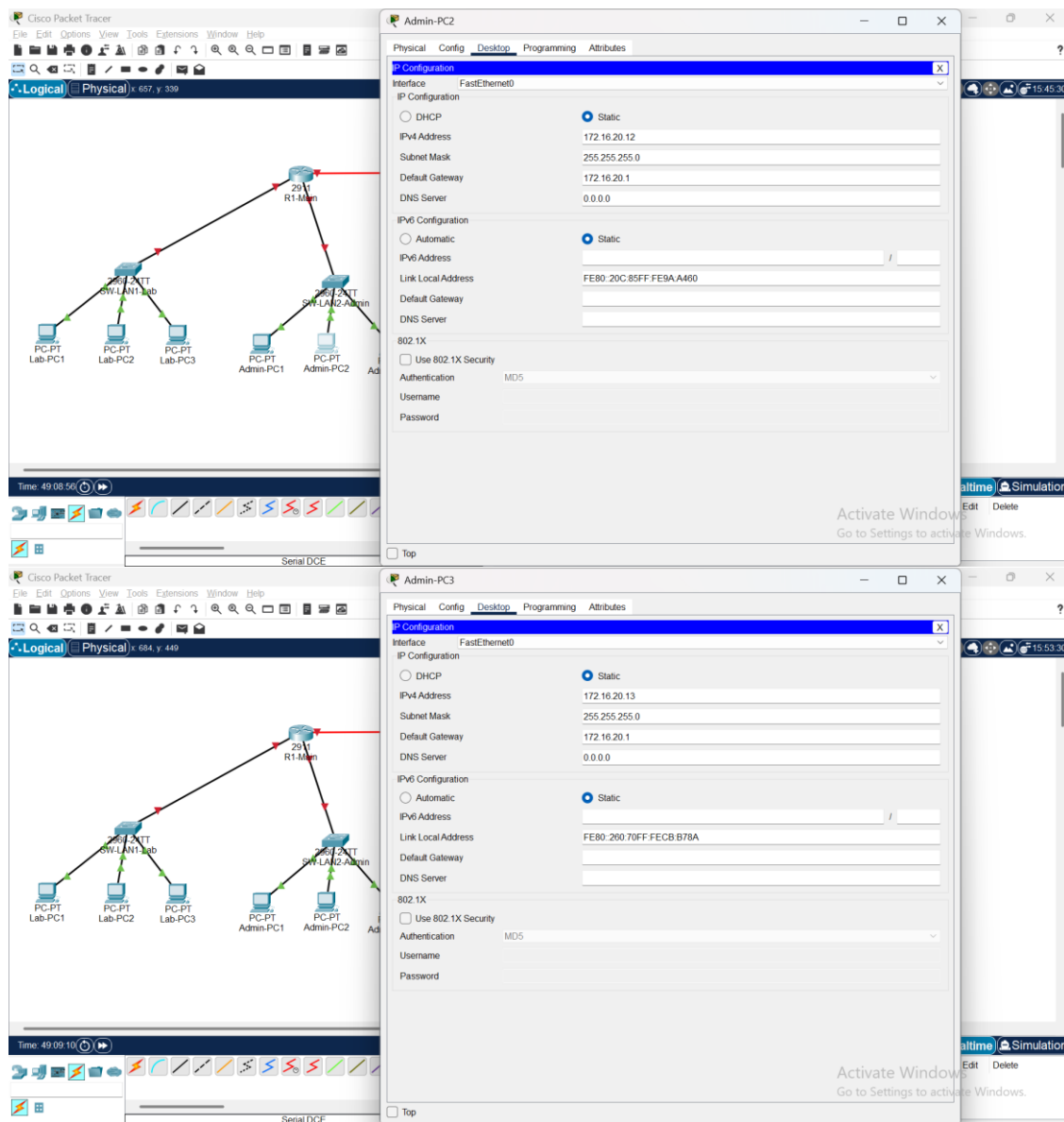


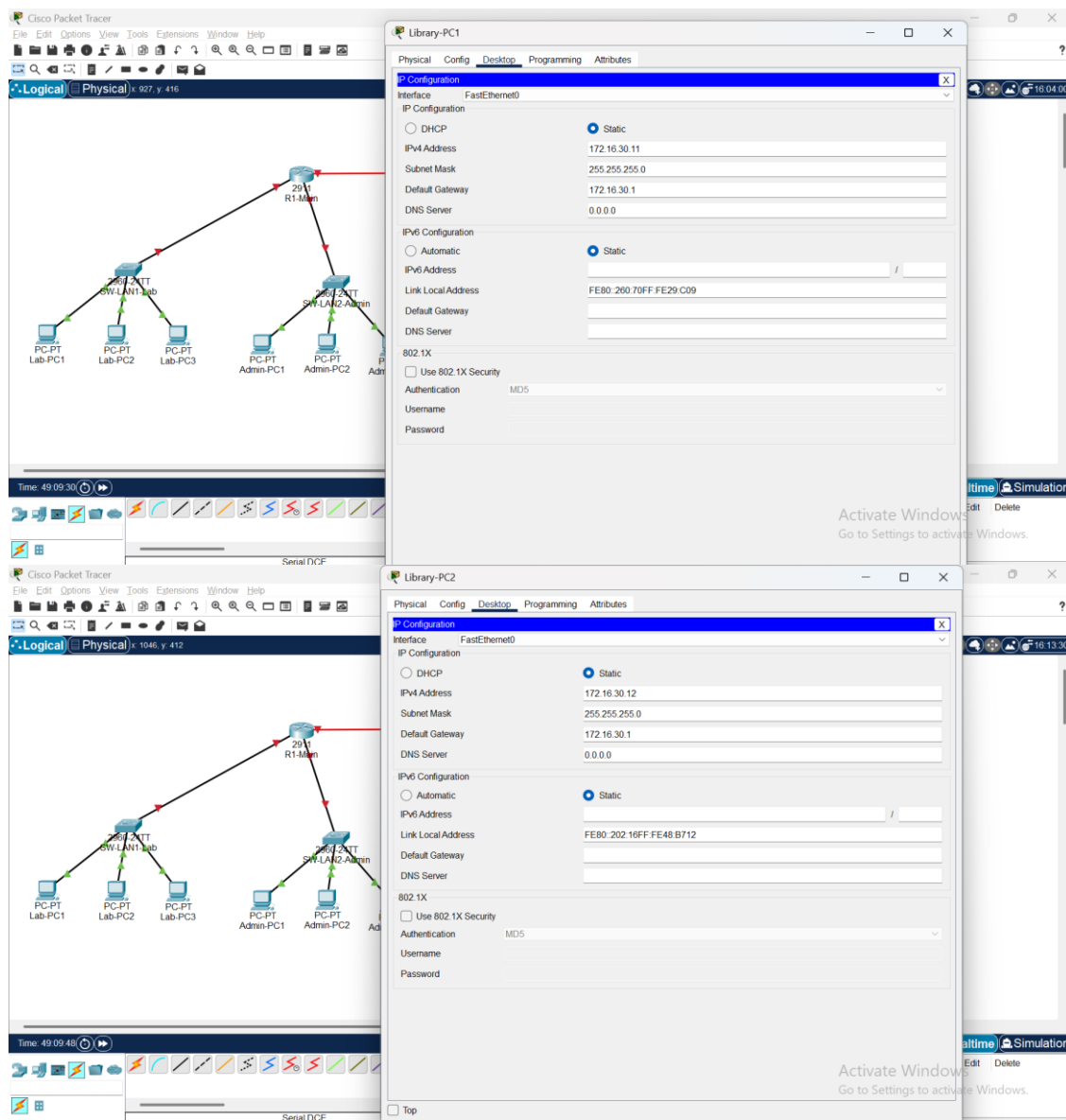


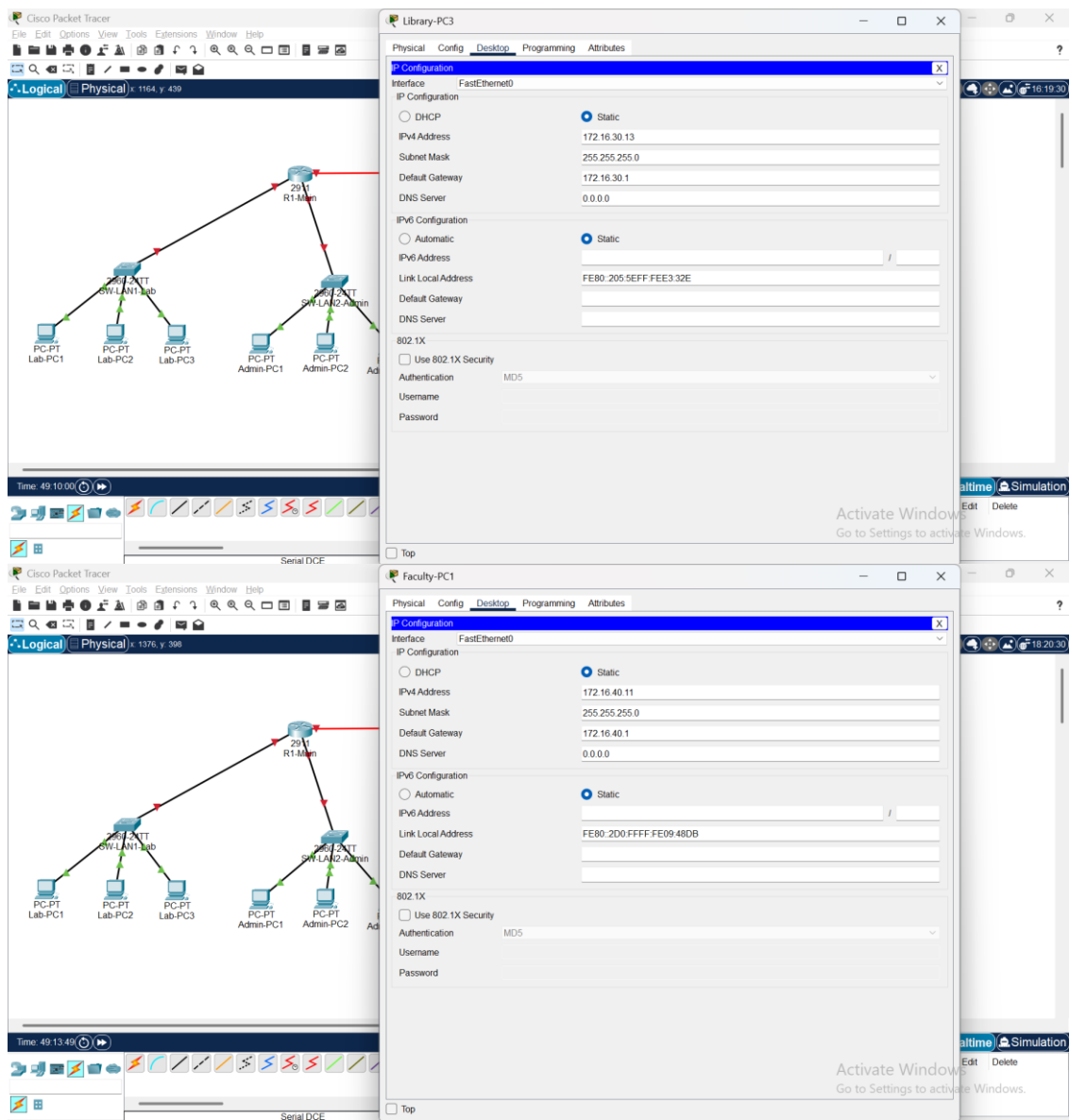


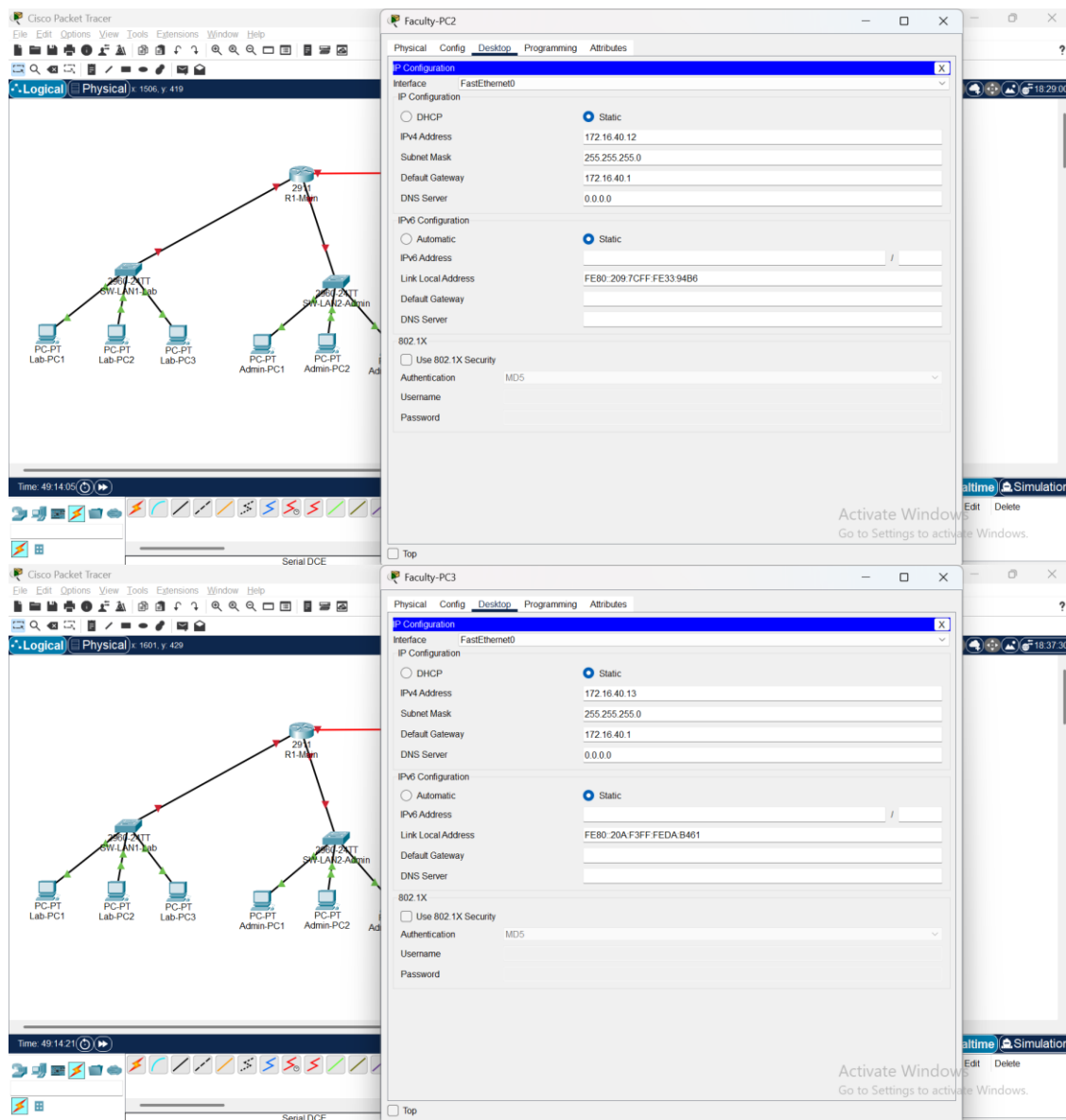


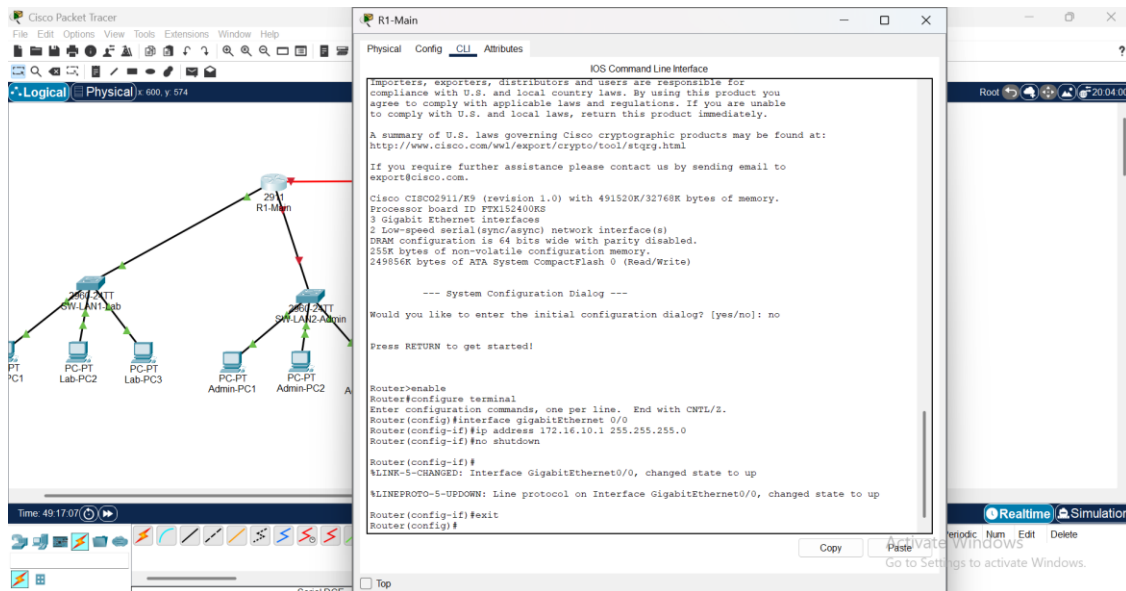












Router>enable

Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface gigabitEthernet 0/0

Router(config-if)#ip address 172.16.10.1 255.255.255.0

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit

Router(config)#

Router#

```

R1-Main
Physical Config CLI Attributes
IOS Command Line Interface

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 172.16.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#interface gigabitEthernet 0/1
Router(config-if)#
% Invalid input detected at '^' marker.

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 172.16.20.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#
  
```

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Top

Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface gigabitEthernet 0/1

Router(config-if)#ip address 172.16.20.1 255.255.255.0

Router(config-if)#no shutdown

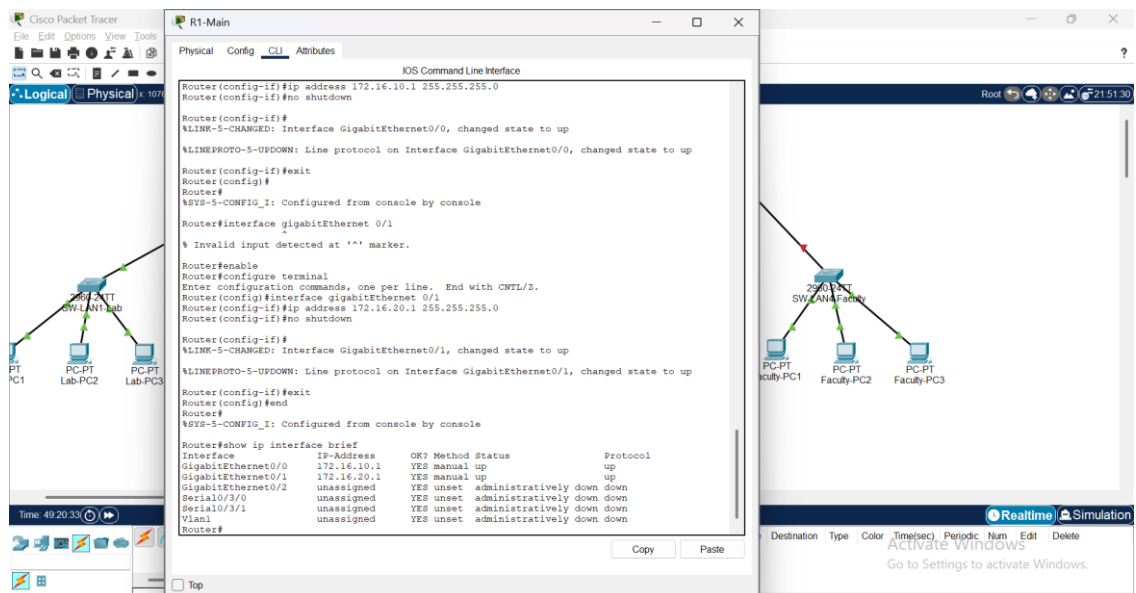
Router(config-if)#

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit

Router(config)#



Router(config)#end

Router#

%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief

Interface IP-Address OK? Method Status Protocol

GigabitEthernet0/0 172.16.10.1 YES manual up up

GigabitEthernet0/1 172.16.20.1 YES manual up up

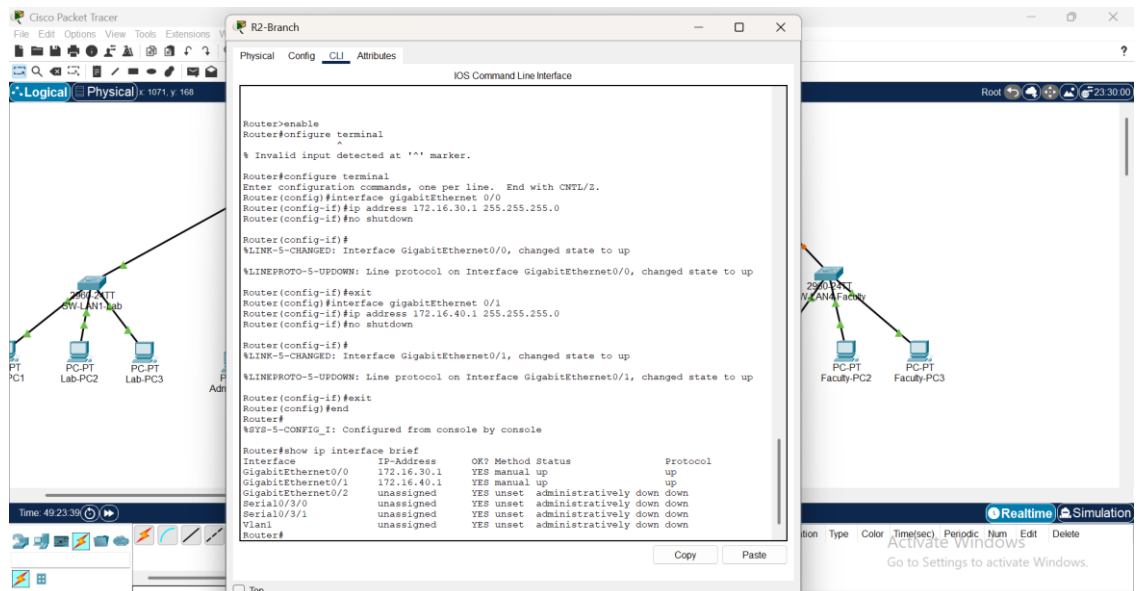
GigabitEthernet0/2 unassigned YES unset administratively down down

Serial0/3/0 unassigned YES unset administratively down down

Serial0/3/1 unassigned YES unset administratively down down

Vlan1 unassigned YES unset administratively down down

Router#



Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface gigabitEthernet 0/0

Router(config-if)#ip address 172.16.30.1 255.255.255.0

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit

Router(config)#interface gigabitEthernet 0/1

Router(config-if)#ip address 172.16.40.1 255.255.255.0

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

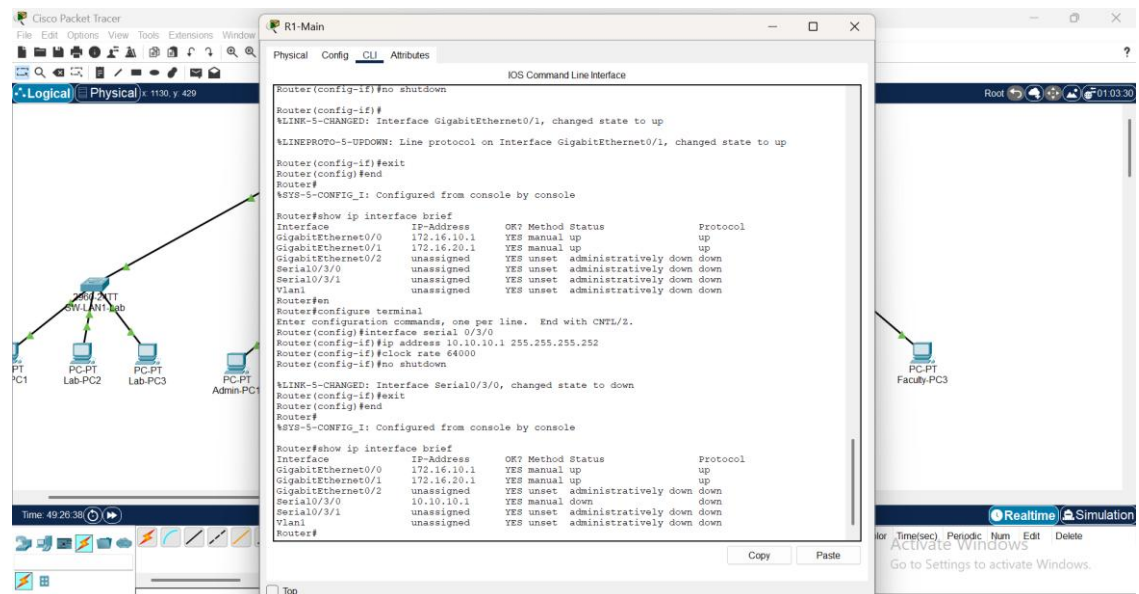
Router(config-if)#exit

Router(config)#end

Router#

%SYS-5-CONFIG_I: Configured from console by console

```
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 172.16.30.1 YES manual up up
GigabitEthernet0/1 172.16.40.1 YES manual up up
GigabitEthernet0/2 unassigned YES unset administratively down down
Serial0/3/0 unassigned YES unset administratively down down
Serial0/3/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router
```



```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface serial 0/3/0
Router(config-if)#ip address 10.10.10.1 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#no shutdown
```

```
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 172.16.10.1 YES manual up up
GigabitEthernet0/1 172.16.20.1 YES manual up up
GigabitEthernet0/2 unassigned YES unset administratively down down
Serial0/3/0 10.10.10.1 YES manual down down
Serial0/3/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#
```

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       172.16.30.1     YES manual  up          up
GigabitEthernet0/1       172.16.40.1     YES manual  up          up
GigabitEthernet0/2       unassigned      YES unset   administratively down down
Serial0/3/0              unassigned      YES unset   administratively down down
Serial0/3/1              unassigned      YES unset   administratively down down
Vlan1                    unassigned      YES unset   administratively down down

Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface serial 0/3/0
Router(config-if)#ip address 10.10.10.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#end
Translating "end"...domain server (255.255.255.255) % Name lookup aborted
Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       172.16.30.1     YES manual  up          up
GigabitEthernet0/1       172.16.40.1     YES manual  up          up
GigabitEthernet0/2       unassigned      YES unset   administratively down down
Serial0/3/0              10.10.10.2      YES manual  up          up
Serial0/3/1              unassigned      YES unset   administratively down down
Vlan1                    unassigned      YES unset   administratively down down

Router#
```

Copy

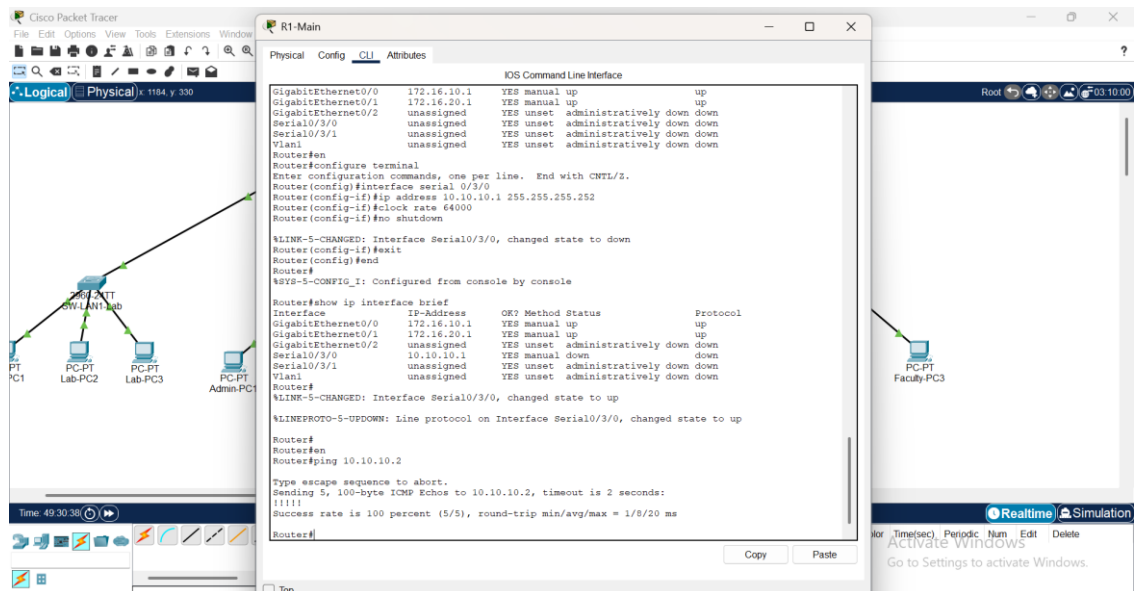
☐ Top

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface serial 0/3/0
Router(config-if)#ip address 10.10.10.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to
up
exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#end
Translating "end"...domain server (255.255.255.255) % Name lookup aborted
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 172.16.30.1 YES manual up up
GigabitEthernet0/1 172.16.40.1 YES manual up up
GigabitEthernet0/2 unassigned YES unset administratively down down
Serial0/3/0 10.10.10.2 YES manual up up
Serial0/3/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#
```



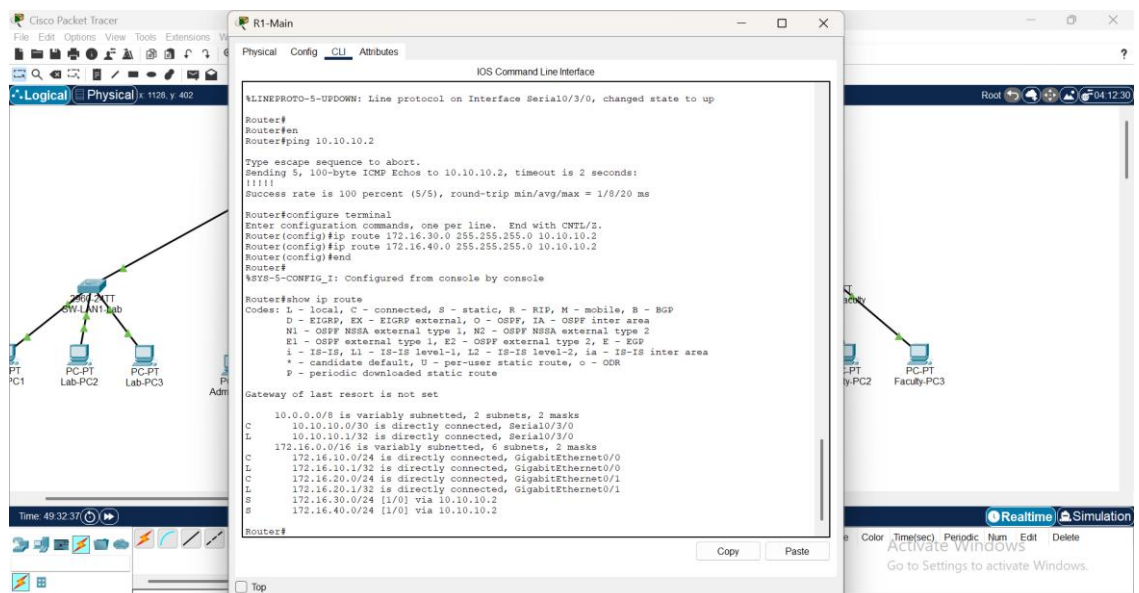
Router#ping 10.10.10.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/8/20 ms



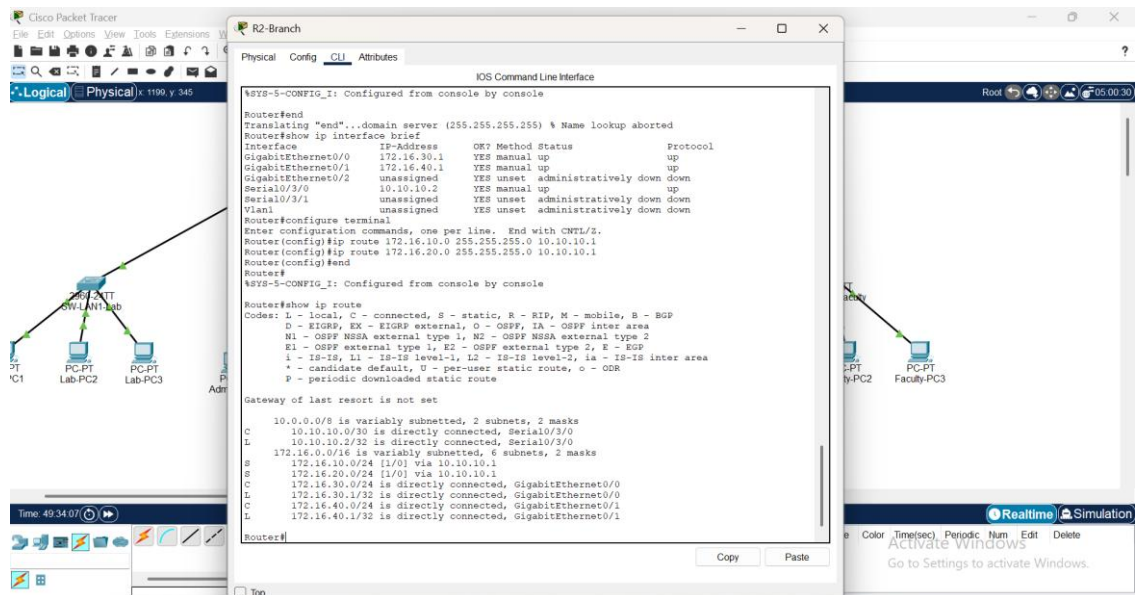
```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 172.16.30.0 255.255.255.0 10.10.10.2
Router(config)#ip route 172.16.40.0 255.255.255.0 10.10.10.2
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
```

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.10.10.0/30 is directly connected, Serial0/3/0
L 10.10.10.1/32 is directly connected, Serial0/3/0
172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks
C 172.16.10.0/24 is directly connected, GigabitEthernet0/0
L 172.16.10.1/32 is directly connected, GigabitEthernet0/0
C 172.16.20.0/24 is directly connected, GigabitEthernet0/1
L 172.16.20.1/32 is directly connected, GigabitEthernet0/1
S 172.16.30.0/24 [1/0] via 10.10.10.2
S 172.16.40.0/24 [1/0] via 10.10.10.2
```

```
Router#
```



Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#ip route 172.16.10.0 255.255.255.0 10.10.10.1

Router(config)#ip route 172.16.20.0 255.255.255.0 10.10.10.1

Router(config)#end

Router#

%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP

i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area

* - candidate default, U - per-user static route, o - ODR

P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

C 10.10.10.0/30 is directly connected, Serial0/3/0

L 10.10.10.2/32 is directly connected, Serial0/3/0

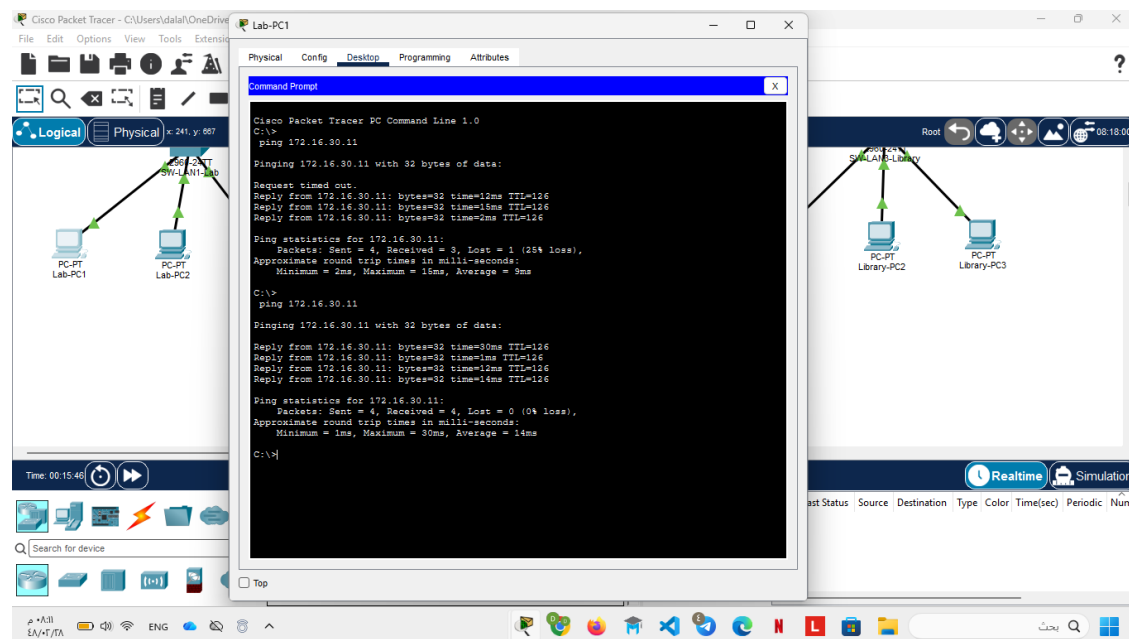
172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks

S 172.16.10.0/24 [1/0] via 10.10.10.1

S 172.16.20.0/24 [1/0] via 10.10.10.1

C 172.16.30.0/24 is directly connected, GigabitEthernet0/0
 L 172.16.30.1/32 is directly connected, GigabitEthernet0/0
 C 172.16.40.0/24 is directly connected, GigabitEthernet0/1
 L 172.16.40.1/32 is directly connected, GigabitEthernet0/1

Router#



```
Cisco Packet Tracer PC Command Line 1.0
C:\>
ping 172.16.30.11

Pinging 172.16.30.11 with 32 bytes of data:

Request timed out.
Reply from 172.16.30.11: bytes=32 time=12ms TTL=126
Reply from 172.16.30.11: bytes=32 time=15ms TTL=126
Reply from 172.16.30.11: bytes=32 time=2ms TTL=126

Ping statistics for 172.16.30.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 15ms, Average = 9ms

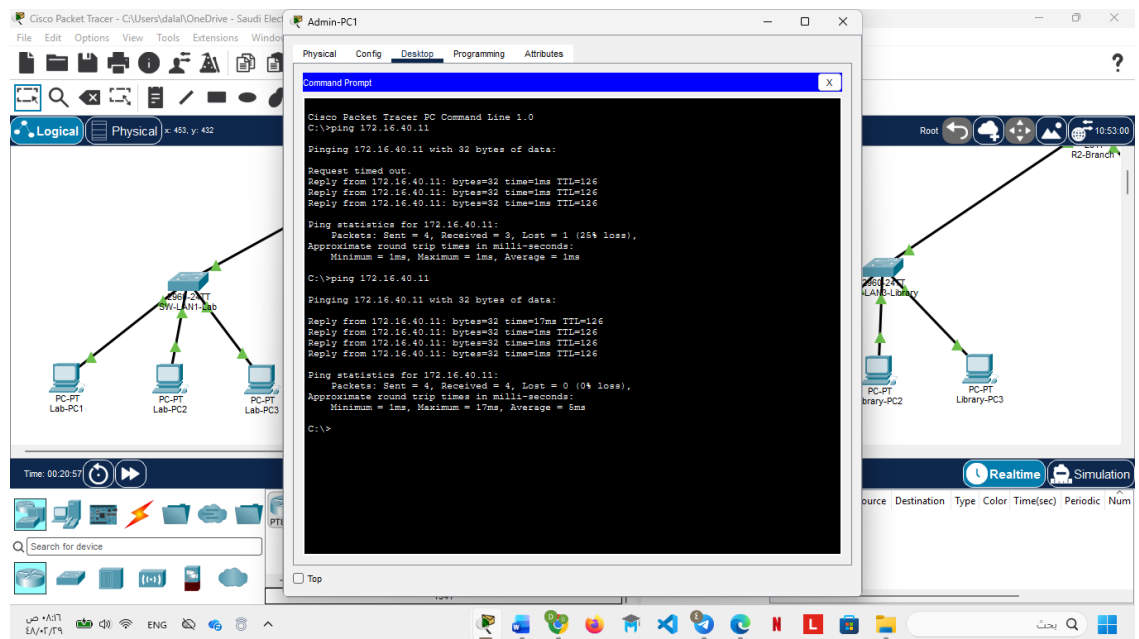
C:\>
ping 172.16.30.11

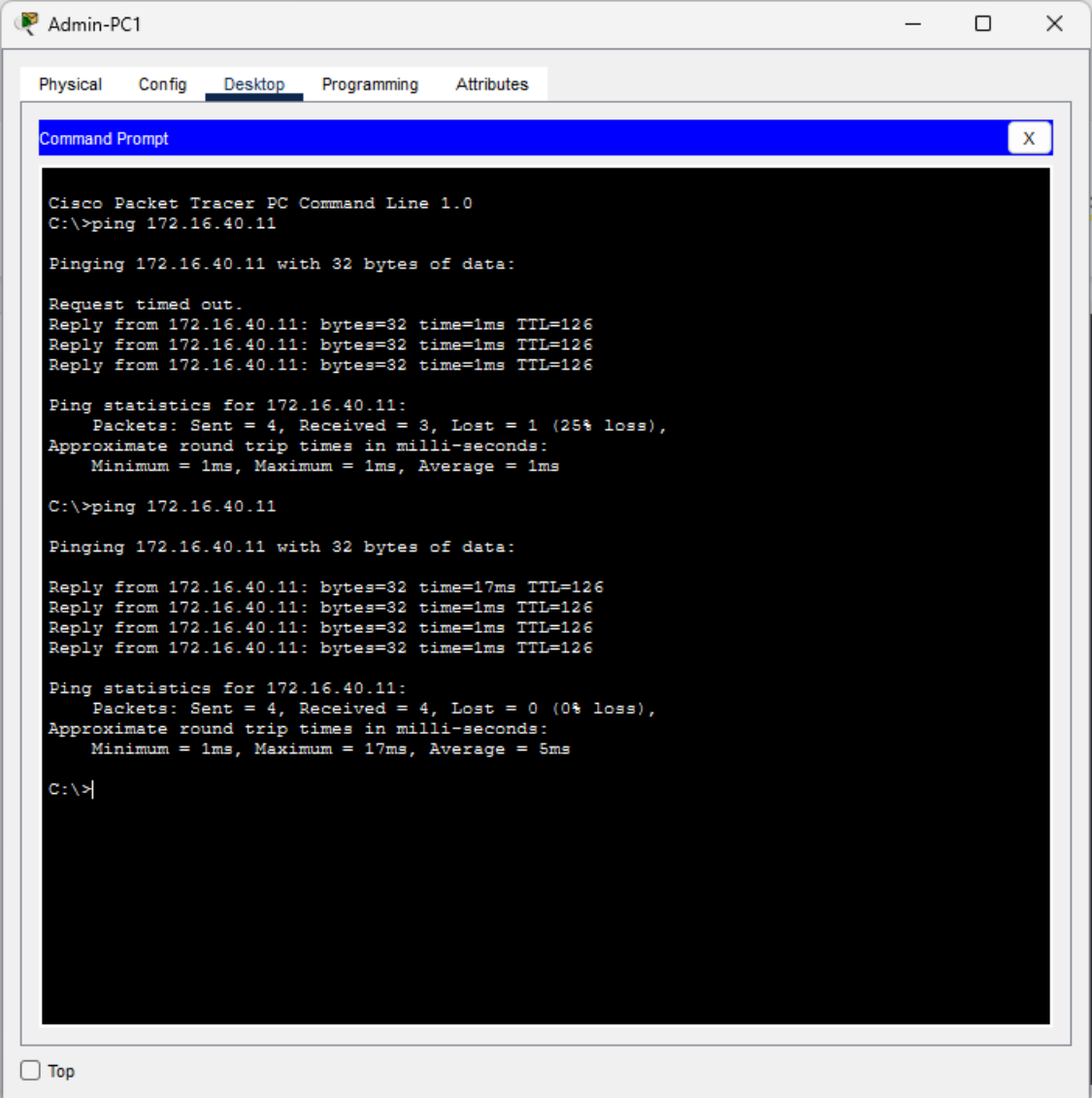
Pinging 172.16.30.11 with 32 bytes of data:

Reply from 172.16.30.11: bytes=32 time=30ms TTL=126
Reply from 172.16.30.11: bytes=32 time=1ms TTL=126
Reply from 172.16.30.11: bytes=32 time=12ms TTL=126
Reply from 172.16.30.11: bytes=32 time=14ms TTL=126

Ping statistics for 172.16.30.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 30ms, Average = 14ms

C:\>
```





The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named 'Admin-PC1'. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Desktop' selected. The command prompt shows the execution of a ping command to 172.16.40.11. The first ping attempt results in a 'Request timed out' and a 25% loss rate. The second ping attempt is successful with a 0% loss rate.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.40.11

Pinging 172.16.40.11 with 32 bytes of data:

Request timed out.
Reply from 172.16.40.11: bytes=32 time=1ms TTL=126
Reply from 172.16.40.11: bytes=32 time=1ms TTL=126
Reply from 172.16.40.11: bytes=32 time=1ms TTL=126

Ping statistics for 172.16.40.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 172.16.40.11

Pinging 172.16.40.11 with 32 bytes of data:

Reply from 172.16.40.11: bytes=32 time=17ms TTL=126
Reply from 172.16.40.11: bytes=32 time=1ms TTL=126
Reply from 172.16.40.11: bytes=32 time=1ms TTL=126
Reply from 172.16.40.11: bytes=32 time=1ms TTL=126

Ping statistics for 172.16.40.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 17ms, Average = 5ms

C:\>|
```

☐ Top

